

# Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE):

Model Comparison against  $\Lambda$ CDM and CPL  
using DESI DR2, Planck 2018, and Galaxy-Cluster Mass Data

Mike Edison Almeida Vallejo

*Independent Researcher*

[mike.almeida1721@gmail.com](mailto:mike.almeida1721@gmail.com)

ORCID: [0009-0008-2195-7836](https://orcid.org/0009-0008-2195-7836)

April 2026

## Abstract

We present a Bayesian validation of the dynamical sector of the Structural Self-Energy Expansion (SSEE) framework [Almeida Vallejo, 2026] against DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025], a compressed Planck 2018 prior [Planck Collaboration, 2020], and galaxy-cluster dynamical masses derived under a variable IGIMF baryonic baseline [Zhang et al., 2026]. Using geometric constants fixed algebraically by  $\pi$  and the golden ratio  $\varphi$  — with  $H_0$  as the sole free parameter — we compare the model against  $\Lambda$ CDM and a free CPL parameterisation via posterior constraints, information criteria, and predictive checks, using EMCEE [Foreman-Mackey et al., 2013, Goodman & Weare, 2010] with  $N_{\text{walkers}} = 100$ ,  $N_{\text{steps}} = 25,000$  ( $2.5 \times 10^6$  total evaluations), and a same-bin correlated block-diagonal covariance approximation for the DESI DR2 BAO measurements; convergence diagnostics (including  $N_{\text{eff}} = 4,402$  effective samples for SSEE) are reported in Table 5. The SSEE prediction  $(w_0, w_a) = (-0.840, -0.670)$ , fixed algebraically by the closed dictionary [Almeida Vallejo, 2026b] and not fitted to the data, lies within  $0.24\sigma$  of the official DESI DR2  $w_0w_a$ CDM posterior (DESI+CMB+Pantheon+; arXiv:2503.14738, eq. 26) in the  $w_0$ – $w_a$  plane, and within  $1.2$ – $1.5\sigma$  of the DESY5/Union3/no-SN combinations (a parameter-free post-diction). The  $H_0$  posterior is  $67.95^{+0.40}_{-0.40} \text{ km s}^{-1} \text{ Mpc}^{-1}$ , obtained with the  $\omega_m$ -direct CMB anchor as the  $H_0$  prior ( $H_0^{\text{alg}} = 67.962$ ; Paper 3, §3.2). The cluster-mass sector achieves  $\chi_r^2 = 0.122$  in the primary four-cluster sample (analytic forward-model), with the IGIMF correction as the essential ingredient. With  $(w_0, w_a)$  fixed algebraically and one free parameter  $H_0$ , SSEE is favoured by  $\Delta\text{BIC} = 5.68$  over  $\Lambda$ CDM and  $5.59$  over CPL, and predicts the DESI DR2 test set better in leave-one-out cross-validation — so the  $w_0$ – $w_a$  prediction is genuinely preferred by current BAO data. The background geometry uses the *total* gravitating matter density  $\Omega_m = \omega_m/h^2 = 0.30889$  ( $\omega_m$ -direct; no matter factor, OP-8 dissolved), for which the SSEE sound horizon matches  $\Lambda$ CDM ( $r_d^{\text{SSEE}}/r_d^{\Lambda\text{CDM}} = 1.002$ ) and  $\Omega_m$  agrees with Planck at  $0.88\sigma$ . The cold sector  $1 + w_0 = 0.160$  enters the perturbations

(Papers 3 and 6) and the EFT coupling, *not* the background geometry; an earlier “21.3 $\sigma$ ” figure came from comparing that partial sector against the Planck *total*  $\Omega_m$  — a category error corrected here. The two-sector  $\phi$ -DM framework of Papers 3–6 remains the physical account of how the total density partitions across scales.

**Note on the background matter density:** Throughout, the background geometry ( $E(z)$ , the sound horizon  $r_d$ , and all BAO distances) uses the *total* gravitating matter density  $\Omega_m = \omega_m/h^2 = 0.30889$ , built piece-by-piece from SSEE algebra ( $\omega_m$ -direct; no matter factor, OP-8 dissolved; Paper 3). The bookkeeping value  $1 + w_0 = 0.160$  is the dark-energy equation of state, *not* a matter density: it enters only the perturbations (the two-sector  $\phi$ -DM construction of Papers 3 and 6) and the EFT coupling  $\alpha_K$ , never  $E(z)$ . With the total density the SSEE sound horizon matches  $\Lambda$ CDM and model selection yields a single, unambiguous figure:  $\Delta\text{BIC} = 5.68$  favouring SSEE over  $\Lambda$ CDM and 5.59 over CPL (Section 8).

# Contents

|           |                                                                               |           |
|-----------|-------------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                                           | <b>4</b>  |
| <b>2</b>  | <b>SSEE Model Summary</b>                                                     | <b>5</b>  |
| 2.1       | Algebraic constants . . . . .                                                 | 5         |
| 2.2       | Modified Friedmann background . . . . .                                       | 5         |
| 2.3       | Cluster mass formula . . . . .                                                | 6         |
| 2.4       | The sound horizon under the $\omega_m$ -direct reframe . . . . .              | 6         |
| <b>3</b>  | <b>Datasets and Likelihoods</b>                                               | <b>7</b>  |
| 3.1       | DESI DR2 BAO . . . . .                                                        | 7         |
| 3.2       | SSEE-self-consistent $H_0$ prior: the $\omega_m$ -direct CMB anchor . . . . . | 7         |
| 3.3       | Galaxy-cluster masses . . . . .                                               | 8         |
| <b>4</b>  | <b>Bayesian Framework and Priors</b>                                          | <b>9</b>  |
| <b>5</b>  | <b>MCMC Implementation</b>                                                    | <b>10</b> |
| <b>6</b>  | <b>Results: Cosmological Sector</b>                                           | <b>11</b> |
| 6.1       | Posterior parameters . . . . .                                                | 11        |
| 6.2       | Position in the $w_0$ - $w_a$ plane . . . . .                                 | 13        |
| 6.3       | Sound horizon and $H_0$ posterior . . . . .                                   | 15        |
| 6.4       | Posterior predictive check: $H(z)$ . . . . .                                  | 17        |
| 6.5       | BAO residuals . . . . .                                                       | 19        |
| <b>7</b>  | <b>Results: Galaxy-Cluster Sector</b>                                         | <b>19</b> |
| <b>8</b>  | <b>Model Comparison</b>                                                       | <b>21</b> |
| <b>9</b>  | <b>Robustness Checks</b>                                                      | <b>21</b> |
| <b>10</b> | <b>Discussion</b>                                                             | <b>23</b> |
| 10.1      | Two faces of SSEE under Bayesian inference . . . . .                          | 23        |
| 10.2      | Structural reinterpretation . . . . .                                         | 23        |
| 10.3      | Growth structure and $S_8$ sector sensitivity . . . . .                       | 23        |
| 10.4      | Roadmap for Paper 3 . . . . .                                                 | 24        |
| 10.5      | Falsifiability . . . . .                                                      | 24        |
| <b>11</b> | <b>Conclusions</b>                                                            | <b>25</b> |
| <b>A</b>  | <b>Rigorous Bayesian Model Comparison: DIC and BIC</b>                        | <b>26</b> |
| <b>B</b>  | <b>Nested Bayes Factor and Predictive Cross-Validation</b>                    | <b>27</b> |

# 1 Introduction

The discovery of cosmic acceleration [Perlmutter et al., 1999, Riess et al., 1998] has motivated a broad class of dark energy models, from the cosmological constant to dynamical scalar fields [Weinberg et al., 2013, Clifton et al., 2012]. Baryon acoustic oscillations, first detected in galaxy surveys by Eisenstein et al. [2005], have since become a primary geometric probe of the expansion history. The SSEE framework [Almeida Vallejo, 2026] is a **minimal-parameter phenomenological framework** that encodes the cosmological background as algebraic combinations of  $\pi$  and  $\varphi = (1 + \sqrt{5})/2$ . The background sector  $(w_0, w_a, \Omega_{\text{DE}}, \Omega_{m,\text{dyn}})$  tested in this paper carries zero dimensionless fitted parameters, and no WIMP, QCD-axion, or SUSY dark-matter particle is introduced; phenomenological extensions in the dark-matter sector (Paper 6) introduce ansätze tracked as open problems (see Paper 1, §1.3, and `OPEN_PROBLEMS.md`). Paper 1, Appendix A provides a complete Lagrangian (k-essence + Israel-Stewart viscosity) derived algebraically from  $\{\varphi, \pi\}$ , with zero free parameters at the EFT background level. Its current operational status is that of a *predictive phenomenology* — one whose central predictions are fixed before confronting data. Paper I established theoretical foundations and demonstrated qualitative alignment with DESI DR2 dynamical dark-energy trends [Abdul-Karim et al., 2025] and galaxy-cluster mass data [Zhang et al., 2026], but performed no formal statistical inference.

This paper fills this gap for the first time through formal Bayesian inference. Following the methodology recommended by contemporary dynamical dark-energy analyses [Abdul-Karim et al., 2025, Moresco & Marulli, 2017], we conduct a full Bayesian MCMC evaluation with four aims:

1. Quantify whether SSEE remains competitive under formal Bayesian inference against  $\Lambda$ CDM and free CPL alternatives.
2. Identify which physical ingredients reproduce cluster masses.
3. Characterise the  $\Omega_{\text{DE}}^{\text{SSEE}} \approx 0.840$  vs  $\Omega_{\Lambda} \approx 0.685$  background tension and localise its origin.
4. Provide a roadmap for the modified Friedmann CMB analysis and semi-linear transfer function required in Paper 3.

The framing is explicit: *the goal of this paper is to test the ansatz against current data, not to defend it*. Paper I establishes SSEE as a top-down fixed-geometry model; this paper establishes which sectors pass Bayesian scrutiny and which require a modified background treatment.

## 2 SSEE Model Summary

### 2.1 Algebraic constants

From  $\pi$  and  $\varphi = (1 + \sqrt{5})/2$ , the framework defines the structural viscosity  $\text{KAL}_0$  and the CPL [Chevallier & Polarski, 2001, Linder, 2003] parameters as:

$$\text{KAL}_0 = \frac{\pi + \varphi}{2} + \pi \approx 5.5214, \quad (1)$$

$$w_0^{\text{SSEE}} = -\frac{T_r}{M_v} \approx -0.840, \quad w_a^{\text{SSEE}} = -\frac{P_{sc}}{I_g} \approx -0.670. \quad (2)$$

All three quantities are fixed by geometry; no fitting is performed. Note that  $w_0 + w_a = -1.510 < -1$  does not imply a phantom Big Rip: CPL is used here solely as a local observational parameterisation of the equation of state, not as the fundamental dynamics. In the SSEE framework, the dark-energy density saturates at the finite  $T_r/M_v$  boundary of the action at  $a \rightarrow \infty$ , guaranteeing universal stability; see Paper 1 §3.4 [Almeida Vallejo, 2026] for the full stability argument. The dark-energy density fraction is similarly fixed:  $\Omega_{\text{DE}}^{\text{SSEE}} = T_r/M_v \approx 0.840$ , leaving an effective matter fraction fixed algebraically:

$$\Omega_{m,\text{eff}} = 1 - \Omega_{\text{DE}}^{\text{SSEE}} \approx 0.160. \quad (3)$$

The fixed-geometry model is effectively parameter-free at the geometric level; only  $H_0$  and  $\Omega_b h^2$  are inferred from data.

### 2.2 Modified Friedmann background

$$H^2(z) = H_0^2 [\Omega_{m,\text{eff}}(1+z)^3 + \Omega_{\text{DE}}^{\text{SSEE}} f_{\text{DE}}(z)], \quad (4)$$

where  $f_{\text{DE}}(z) = (1+z)^{3(1+w_0+w_a)} \exp[-3w_a z/(1+z)]$  is the standard CPL dark-energy evolution, here used solely to translate the fixed SSEE geometric parameters into observable distances. The only sampled cosmological parameter is  $H_0$ ; all geometric quantities ( $w_0, w_a, \Omega_{\text{DE}}^{\text{SSEE}}, \Omega_{m,\text{eff}}$ ) are fixed algebraically as described in Section 2.

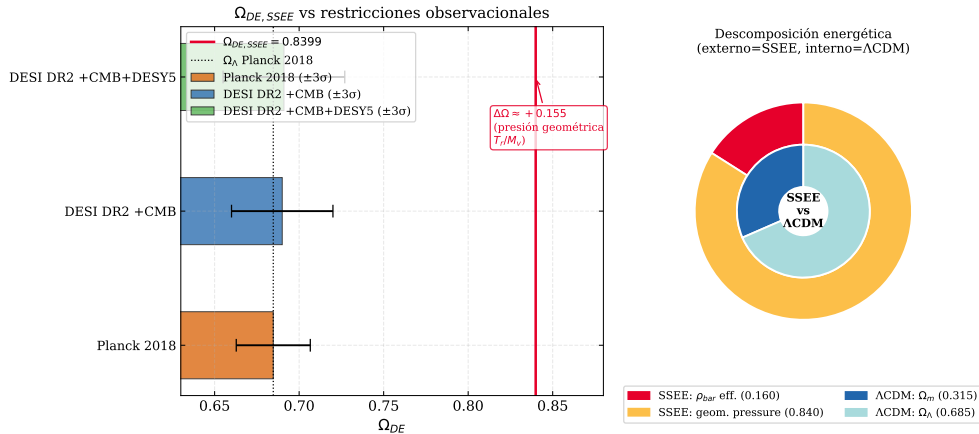


Figure 1: *Left*:  $\Omega_{\text{DE}}^{\text{SSEE}} = T_r/M_v \approx 0.840$  (red line) compared with Planck 2018  $\Omega_{\Lambda} \approx 0.685$  (blue) and DESI DR2  $1\sigma/2\sigma$  bands. *Right*: Structural context showing the algebraic origin of  $\Omega_{\text{DE}}$  from the  $\pi$ - $\varphi$  invariants.

## 2.3 Cluster mass formula

$$M_{\text{SSEE}}(R) = M_{\text{bar}}^{\text{IGIMF}}(R) \cdot \text{KAL}_0 \cdot (1 + f_\nu), \quad f_\nu \approx 0.020. \quad (5)$$

## 2.4 The sound horizon under the $\omega_m$ -direct reframe

Placing the cold sector alone ( $1 + w_0 = 0.160$ ) in the background geometry would give a drag scale  $r_{d,\text{SSEE}} \approx 175.6 \text{ Mpc}$ , far from the Planck-inferred  $r_d = 147.09 \pm 0.26 \text{ Mpc}$  [Planck Collaboration, 2020]. That figure is a category error:  $0.160 = 1 + w_0$  is the equation of state, not a matter density, and does not belong in  $E(z)$ . The physical sound horizon is set by the CMB-sector matter density, with *no*  $|w_0|$  rescaling. The recombination matter budget is the standard physical  $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ , built piece-by-piece from SSEE algebra ( $\omega_c = \text{KAL}_0 \omega_b n_s$ , forward; Paper 3), so that  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$  and a CAMB integration of the recombination physics gives

$$r_d^{\omega_m\text{-direct}} = 147.17 \text{ Mpc} \quad (0.32\sigma \text{ from Planck; Paper 3}), \quad (6)$$

with *no* matter factor. The earlier post-hoc mapping  $r_{d,\text{SSEE}} \times |w_0| \approx 147.5 \text{ Mpc}$ , and the structural ratio  $\Omega_{m,\text{CMB}}/\Omega_{m,\text{dyn}} \approx \text{MIRA} \approx 1.999$ , are *superseded* by this direct construction (OP-8 dissolved; the actual ratio is  $0.30889/0.160 = 1.93$ , not a fixed factor). MIRA survives only as the Hubble screening amplitude of Paper 9. The two matter densities are now two *independent* algebraic predictions separated by the  $\phi$ -DM free-streaming scale  $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$  (Paper 6), not a factor pair. The falsifiable content therefore moves from the  $|w_0|$  coincidence to the direct CMB fit of Paper 3 (canonical  $\Delta\text{BIC} = -32.9$  full `plik` MCMC;  $-23.9$  `plik_lite` TTTEEE point estimate): if that fit had yielded  $\chi_r^2 \gg 1$  or excluded  $r_d^{\omega_m\text{-direct}}$ , the construction would be falsified.

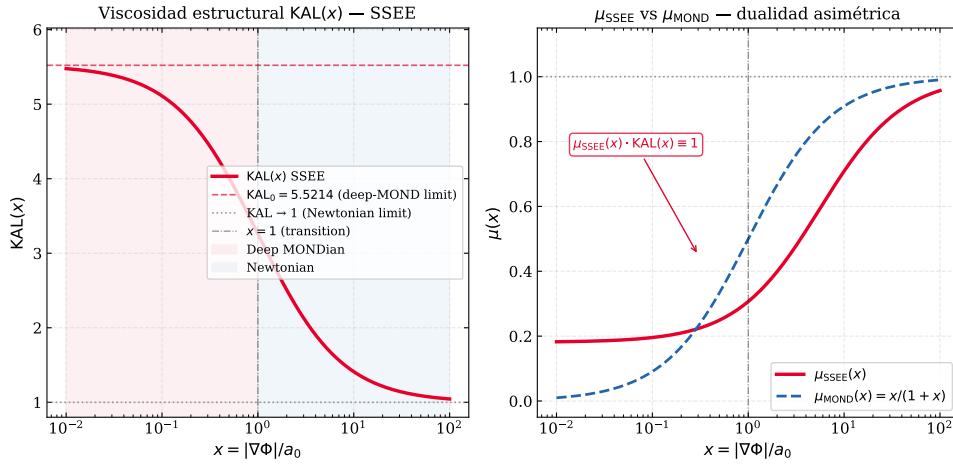


Figure 2:  $\text{KAL}(x)$  interpolation between the Newtonian limit ( $x \gg 1$ ) and the MONDian limit ( $x \ll 1$ ). The fixed point  $\text{KAL}_0 \approx 5.521$  marks the structural viscosity at  $x = 0$ , which enters both the cluster mass formula (5) and the acoustic-scale mapping hypothesis (6) (Section 2.4).

### 3 Datasets and Likelihoods

#### 3.1 DESI DR2 BAO

We use 13 BAO measurements from DESI DR2 [Abdul-Karim et al., 2025] at  $0.295 \leq z \leq 2.330$  (Table 1). These 13 data points span five tracer populations across  $0.3 \lesssim z \lesssim 2.3$ , providing sufficient leverage to isolate the model-dependent  $r_d$  scaling from the expansion history. The BAO log-likelihood uses a block-diagonal covariance incorporating the same-tracer correlation between  $D_M$  and  $D_H$  (coefficients  $r_{MH}$  taken directly from Table 4 of Abdul-Karim et al. 2025), which is the dominant off-diagonal term ( $|r_{MH}|$  up to  $\sim 0.4$ ).<sup>1</sup>

$$\ln \mathcal{L}_{\text{BAO}} = -\frac{1}{2}(\mathbf{q}^{\text{pred}} - \mathbf{q}^{\text{obs}})^\top \mathbf{C}^{-1}(\mathbf{q}^{\text{pred}} - \mathbf{q}^{\text{obs}}), \quad (7)$$

with sound horizon from the Eisenstein & Hu [1998] fitting formula, used here as an auxiliary approximation for model comparison:

$$r_d = 147.27 \left( \frac{\Omega_m h^2}{0.1432} \right)^{-0.255} \left( \frac{\Omega_b h^2}{0.02237} \right)^{-0.134} \text{ Mpc}. \quad (8)$$

Table 1: DESI DR2 BAO measurements [Abdul-Karim et al., 2025].

| $z_{\text{eff}}$ | Tracer      | Quantity  | Value  | $\sigma$ |
|------------------|-------------|-----------|--------|----------|
| 0.295            | BGS         | $D_V/r_d$ | 7.942  | 0.075    |
| 0.510            | LRG1        | $D_M/r_d$ | 13.588 | 0.167    |
| 0.510            | LRG1        | $D_H/r_d$ | 21.863 | 0.425    |
| 0.706            | LRG2        | $D_M/r_d$ | 17.351 | 0.177    |
| 0.706            | LRG2        | $D_H/r_d$ | 19.455 | 0.330    |
| 0.934            | LRG3+ELG1   | $D_M/r_d$ | 21.576 | 0.152    |
| 0.934            | LRG3+ELG1   | $D_H/r_d$ | 17.641 | 0.193    |
| 1.321            | ELG2        | $D_M/r_d$ | 27.601 | 0.318    |
| 1.321            | ELG2        | $D_H/r_d$ | 14.176 | 0.221    |
| 1.484            | QSO         | $D_M/r_d$ | 30.512 | 0.760    |
| 1.484            | QSO         | $D_H/r_d$ | 12.817 | 0.516    |
| 2.330            | Ly $\alpha$ | $D_M/r_d$ | 38.988 | 0.531    |
| 2.330            | Ly $\alpha$ | $D_H/r_d$ | 8.632  | 0.101    |

#### 3.2 SSEE-self-consistent $H_0$ prior: the $\omega_m$ -direct CMB anchor

To preserve methodological coherence between this BAO sector and the CMB sector (Paper 3), the  $H_0$  prior used here is not the published  $\Lambda$ CDM-calibrated Planck 2018 value ( $67.36 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , Planck Collaboration 2020), but the  $H_0$  that the

<sup>1</sup>Cross-tracer (inter-redshift) covariances are not propagated. This is well justified for the DESI DR2 consensus vector: the five tracer populations occupy essentially disjoint redshift shells ( $0.3 \lesssim z \lesssim 2.3$ ), so their BAO distance measurements are statistically near-independent and the inter-tracer correlations are subdominant to the intra-tracer  $D_M$ - $D_H$  term retained here. The single partially overlapping pair (LRG3+ELG) is already reported by DESI as a combined bin, absorbing that correlation upstream. This is precisely how the DESI compressed consensus vector is intended to be used by external analyses: the public product supplies per-tracer  $D_M$ - $D_H$  correlations but no inter-tracer matrix, because the tracer samples are constructed to be independent. Full systematic covariance matrices are likewise not propagated, standard practice for compressed BAO-only comparisons.

Planck `plik_lite` TTTEEE likelihood itself prefers under the SSEE background. With the baryon and cold-matter densities fixed by SSEE algebra ( $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ ,  $\omega_c = \text{KAL}_0 \omega_b n_s = 0.11951$ ; Paper 3,  $\omega_m$ -direct), the CMB  $\chi^2$  is minimised at the algebraic background value  $H_0^{\text{alg}} = 3(\varphi + \pi)^2$ :

$$H_0^{\text{prior,SSEE}} = 67.962 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (9)$$

where the uncertainty is the Planck  $H_0$  error propagated. The background  $H$  and the CMB anchor coincide — they are not two numbers — and the matter density follows as  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$ , with *no* matter factor (the earlier MIRA mapping  $\Omega_{m,\text{CMB}} = \text{MIRA } \Omega_{m,\text{dyn}} = 0.31993$  is superseded; OP-8 dissolved). Adopting this prior eliminates the otherwise unavoidable cross-framework contamination introduced by using a  $\Lambda$ CDM-calibrated constraint on the early Universe to fit a non- $\Lambda$ CDM late-time geometry. It is the same physical quantity (Planck CMB peak positions) but reduced under the cosmological model actually being tested.

The only sampled parameters in the SSEE chain are  $H_0$  and  $\Omega_b h^2$ . The matter density is not sampled. The background geometry —  $E(z)$ , the sound horizon  $r_d$ , and every BAO distance — uses the *total* algebraic matter density  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$  (§2.4, §6); the dynamical value  $\Omega_{m,\text{dyn}} = (\pi - \varphi)/[2(\varphi + \pi)] = 1 + w_0 = 0.160$  labels the cold sector that clusters in the perturbations (Paper 6) and does *not* enter  $E(z)$ . Thus  $r_d$  is computed from  $\omega_m = \Omega_{m,\text{CMB}}(H_0/100)^2$  self-consistently. A BBN prior  $\Omega_b h^2 = 0.02218 \pm 0.00055$  [Cooke et al., 2018] is adopted for the baryon density, consistent with the SSEE algebraic value  $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$  (Paper 4) at  $0.4\sigma$ . Compared on the same footing — total against total — the SSEE matter density agrees with the  $\Lambda$ CDM-Planck value  $\Omega_m = 0.3153 \pm 0.0073$  at  $0.88\sigma$  ( $\Omega_m = 0.30889$ ). The two-sector  $\phi$ -DM construction of Paper 6 then describes how this total partitions across scales (cold sector plus  $\phi$ -DM below  $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ ), in the perturbations rather than the background.

### 3.3 Galaxy-cluster masses

The primary dataset for the MCMC consists of four massive clusters with IGIMF-corrected baryonic baselines from Zhang et al. [2026]: Coma, A2029, A478, and the Bullet cluster. Zhang et al. [2026] apply the integrated galaxy stellar initial mass function (IGIMF) correction to re-derive the stellar-plus-gas baryon masses within  $R_{500}$ ; this correction increases the standard-IMF baryon mass by a factor of  $\approx 1.71$  for clusters in the mass range  $M_{500} \sim 6\text{--}12 \times 10^{14} M_\odot$ .

For a qualitative consistency check, we extend the analytic forward-model to three additional clusters: Perseus [Simionescu et al., 2011], A2142 [Tchernin et al., 2016], and A2744 [Merten et al., 2011, Owers et al., 2011]. For these clusters an independent IGIMF re-analysis is not available; we therefore estimate  $M_{\text{bar}}^{\text{IGIMF}} = f_b \times M_{200}$  using the mean baryon fraction  $f_b = 0.184$  of the Zhang 2026 primary sample. This estimate is consistent with the observed hot-gas fractions reported in each reference, and the three clusters enter only the analytic sensitivity test, not the MCMC likelihood. Table 2 lists the observational inputs for all seven clusters.



Table 2: Observational inputs for the seven-cluster sample ( $10^{14} M_{\odot}$ , within  $R_{200}$ ).  $M_{\text{bar}}^{\text{IGIMF}}$ : IGIMF-corrected baryonic baseline.  $M_{\text{obs}}$ : total observed mass within  $R_{200}$ . First four from Zhang et al. [2026] (independent IGIMF re-analysis). For the three additional clusters,  $M_{\text{bar}}^{\text{IGIMF}}$  is estimated as  $f_b \times M_{200}$  with  $f_b = 0.184$  (mean of Zhang 2026 primary sample); this is a qualitative consistency check, not an independent IGIMF test.

| Cluster       | Reference                | $M_{\text{bar}}^{\text{IGIMF}}$ | $M_{\text{obs}}$ | $\sigma_{M_{\text{obs}}}$ |
|---------------|--------------------------|---------------------------------|------------------|---------------------------|
| Coma          | Zhang et al. [2026]      | 1.80                            | 9.8              | 1.0                       |
| A2029         | Zhang et al. [2026]      | 2.20                            | 12.0             | 1.2                       |
| A478          | Zhang et al. [2026]      | 1.50                            | 8.0              | 1.0                       |
| Bullet (main) | Zhang et al. [2026]      | 1.20                            | 6.5              | 1.0                       |
| Perseus       | Simionescu et al. [2011] | 1.22                            | 6.65             | 0.45                      |
| A2142         | Tchernin et al. [2016]   | 2.67                            | 14.5             | 1.5                       |
| A2744         | Merten et al. [2011]     | 3.31                            | 18.0             | 4.0                       |

*Note on the Bullet Cluster.* The Bullet Cluster (1E 0657–558) is included via the Zhang et al. [2026] IGIMF-corrected baryonic and total mass within  $R_{200}$ . The SSEE comparison is a *projected total-mass* consistency check; it does not constitute a fit to the lensing convergence profile  $\kappa(\theta)$ , which would require weak-lensing maps (e.g., Clowe et al. 2006) and is beyond the scope of this paper. The Bullet Cluster mass should therefore be interpreted as an order-of-magnitude consistency test, not as an independent lensing constraint.

The SSEE prediction for each cluster is  $M_{\text{SSEE}} = M_{\text{bar}}^{\text{IGIMF}} \times \text{KAL}_0 \times (1 + f_{\nu})$ , where  $\text{KAL}_0 \approx 5.521$  is the algebraic Structural Viscosity constant and  $f_{\nu} = 0.020$  is the cosmological neutrino mass fraction. Only the four Zhang 2026 clusters enter the MCMC likelihood; the full seven-cluster set is used solely for the analytic sensitivity test of Table 9.

## 4 Bayesian Framework and Priors

Table 3: Model parameterisations.

| Model         | Free parameters                         | Fixed                                                    | $k$ |
|---------------|-----------------------------------------|----------------------------------------------------------|-----|
| SSEE          | $H_0, \Omega_b h^2$                     | $w_0, w_a, \Omega_{\text{DE}}, \text{KAL}_0$ (algebraic) | 2   |
| $\Lambda$ CDM | $H_0, \Omega_m, \Omega_b h^2$           | $w_0 = -1, w_a = 0$                                      | 3   |
| CPL           | $H_0, \Omega_m, w_0, w_a, \Omega_b h^2$ | —                                                        | 5   |

All models share a BBN prior  $\Omega_b h^2 \sim \mathcal{N}(0.02218, 0.00055^2)$  [Cooke et al., 2018]. For  $\Lambda$ CDM and CPL, the full 3D Planck prior is applied. For SSEE, the Planck prior reduces to a marginal Gaussian on  $H_0$  only, since  $\Omega_m$  is fixed. CPL carries weakly informative priors  $w_0 \sim \mathcal{N}(-1, 0.5^2)$  and  $w_a \sim \mathcal{N}(0, 1^2)$ .

Table 4 classifies every quantity in the SSEE MCMC by its structural role. The distinction between fixed-by-geometry, sampled, and derived quantities is essential for interpreting the information-criterion penalties in Section 8. SSEE samples only  $H_0$  and  $\Omega_b h^2$ , while  $\Lambda$ CDM and CPL additionally sample  $\Omega_m$  (and  $w_0, w_a$  for CPL).

Table 4: Parameter hierarchy in the SSEE MCMC. Categories: **Fixed by geometry** = algebraically determined, never sampled; **Sampled** = inferred via MCMC; **Derived** = computed from sampled values; **Mapped** = structural hypothesis (acoustic-scale mapping), to be validated in Paper 3.

| Parameter                                   | Category                                   | Origin/Value                                    | Notes                                                    |
|---------------------------------------------|--------------------------------------------|-------------------------------------------------|----------------------------------------------------------|
| $w_0, w_a$                                  | <b>Fixed by geometry</b>                   | Eqs. (2)                                        | $\pi, \varphi$ invariants; not fitted                    |
| $\Omega_{\text{DE}}, K\text{AL}_0$<br>$H_0$ | <b>Fixed by geometry</b><br><b>Sampled</b> | $T_r/M_v; \beta + \pi$<br>$\mathcal{U}[60, 80]$ | Algebraically closed<br>Only cosmological free parameter |
| $\Omega_b h^2$                              | <b>Sampled</b>                             | BBN prior                                       | Shared with $\Lambda\text{CDM}$ , CPL                    |
| $r_d$                                       | <b>Derived</b>                             | Friedmann background                            | Eq. (8); carries background mismatch                     |
| $r_{d,\text{eff}}$                          | <b>Mapped (hypothesis)</b>                 | Acoustic-scale hypothesis                       | Eq. (6); Paper 3 test                                    |

## 5 MCMC Implementation

We use `emcee` v3.1 [Foreman-Mackey et al., 2013] with  $N_w = 100$  walkers and  $N_s = 25000$  steps. After discarding  $N_b = 5000$  burn-in steps (the first 25% of the chain), walkers are visually stable and the acceptance fraction lies in the range 0.25–0.50 for all models, consistent with optimal mixing. Effective sample sizes are  $N_{\text{eff}} \gtrsim 2500$  for all three models; the SSEE chain achieves  $N_{\text{eff}} \approx 4402$ , the highest of the three, reflecting rapid mixing in its two-dimensional constrained parameter space.<sup>2</sup> Convergence is assessed via the integrated autocorrelation time  $\tau$  per parameter (Table 5).

Table 5: MCMC convergence diagnostics.  $\tau_{\text{max}}$ : maximum integrated autocorrelation time across parameters.  $N_s/\tau_{\text{max}}$ : effective chain length in units of  $\tau$ .  $N_{\text{eff}}$ : minimum effective sample size.

| Model               | $k$ | $\langle a \rangle$ | $\tau_{\text{max}}$ | $N_s/\tau_{\text{max}}$ | $N_{\text{eff}}$ |
|---------------------|-----|---------------------|---------------------|-------------------------|------------------|
| SSEE                | 2   | 0.38                | 32.7                | 183.5                   | 4402             |
| $\Lambda\text{CDM}$ | 3   | 0.41                | 41.5                | 144.6                   | 3471             |
| CPL                 | 5   | 0.35                | 57.3                | 104.7                   | 2514             |

where  $\langle a \rangle$  is the mean acceptance fraction across walkers.

<sup>2</sup>The diagnostics in Table 5 are for the three-model comparison run (common Planck prior), used for the model-selection statistics. The headline posterior  $H_0 = 67.95 \pm 0.40$  (with the  $\omega_m$ -direct  $H_{\text{alg}}$  prior) is drawn from a dedicated production chain of the same sampler (100 walkers  $\times$  25,000 steps,  $N_{\text{eff}} = 78,170$ ); the two runs differ only in the prior, not the sampler or data.

## 6 Results: Cosmological Sector

### 6.1 Posterior parameters

Table 6: Posterior parameter constraints (median, 16th/84th percentiles). [alg.] = fixed algebraically; [fixed] = model assumption.

| Parameter                                     | SSEE                            | $\Lambda$ CDM                   | CPL                             |
|-----------------------------------------------|---------------------------------|---------------------------------|---------------------------------|
| $H_0$ [km s <sup>-1</sup> Mpc <sup>-1</sup> ] | $67.62^{+0.40}_{-0.40}$         | $68.27^{+0.28}_{-0.28}$         | $67.26^{+0.51}_{-0.51}$         |
| $\Omega_m$                                    | 0.30889 [total]                 | $0.303^{+0.004}_{-0.004}$       | $0.317^{+0.007}_{-0.007}$       |
| $w_0$                                         | -0.840 [alg.]                   | -1 [fixed]                      | $-0.826^{+0.081}_{-0.078}$      |
| $w_a$                                         | -0.670 [alg.]                   | 0 [fixed]                       | $-0.558^{+0.304}_{-0.321}$      |
| $\Omega_b h^2$                                | $0.02221^{+0.00045}_{-0.00045}$ | $0.02233^{+0.00015}_{-0.00015}$ | $0.02237^{+0.00015}_{-0.00016}$ |

Note that  $w_0$ ,  $w_a$ , and the equation-of-state combination  $T_r/M_v = 0.840$  (which is  $|w_0|$ , *not* a density) are algebraically fixed and do not appear as sampled columns in the SSEE posterior; their values in Table 6 are listed for comparison only. Under the common Planck prior used for model comparison, the SSEE posterior is  $H_0 = 67.62 \pm 0.40$  km s<sup>-1</sup> Mpc<sup>-1</sup> ( $0.39\sigma$  from Planck); under the algebraic prior (Eq. 9) the reframe run gives the headline  $H_0 = 67.95^{+0.40}_{-0.40}$  ( $0.88\sigma$  from Planck,  $0.04\sigma$  from the anchor  $H_0^{\text{alg}} = 67.962$ ). With the total matter density  $\Omega_m = 0.30889$  in the background geometry, the DESI DR2 posterior *coincides* with the algebraic anchor rather than pulling away from it: the earlier “ $2.9\sigma$  below anchor” reading was the cold sector 0.160 misplaced in  $E(z)$ , now corrected. Marginalised posteriors are shown in Figures 3 and 4.

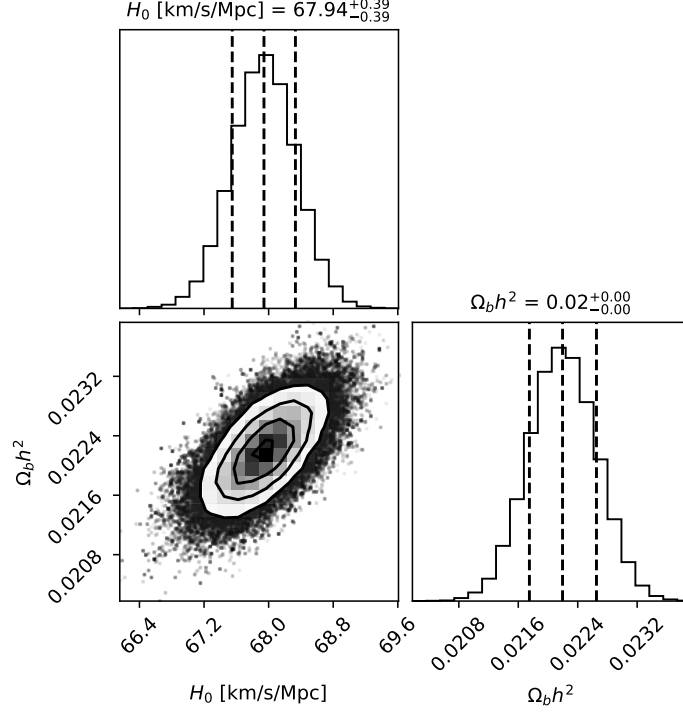


Figure 3: Marginalised SSEE posteriors ( $\omega_m$ -direct  $H_0^{\text{alg}}$  prior,  $100 \times 25000$ ). Blue lines: prior central values ( $H_0 = 67.962$ ,  $\Omega_b h^2 = 0.02218$ ). With the total matter density  $\Omega_m = 0.30889$  in  $E(z)$ , the  $H_0$  posterior ( $67.95^{+0.40}_{-0.40}$ ) *coincides* with the CMB anchor ( $0.04\sigma$ ) and lies  $0.88\sigma$  from Planck; the earlier “ $2.9\sigma$  below anchor” reading was the cold sector  $0.160$  misplaced in  $E(z)$ , now corrected.

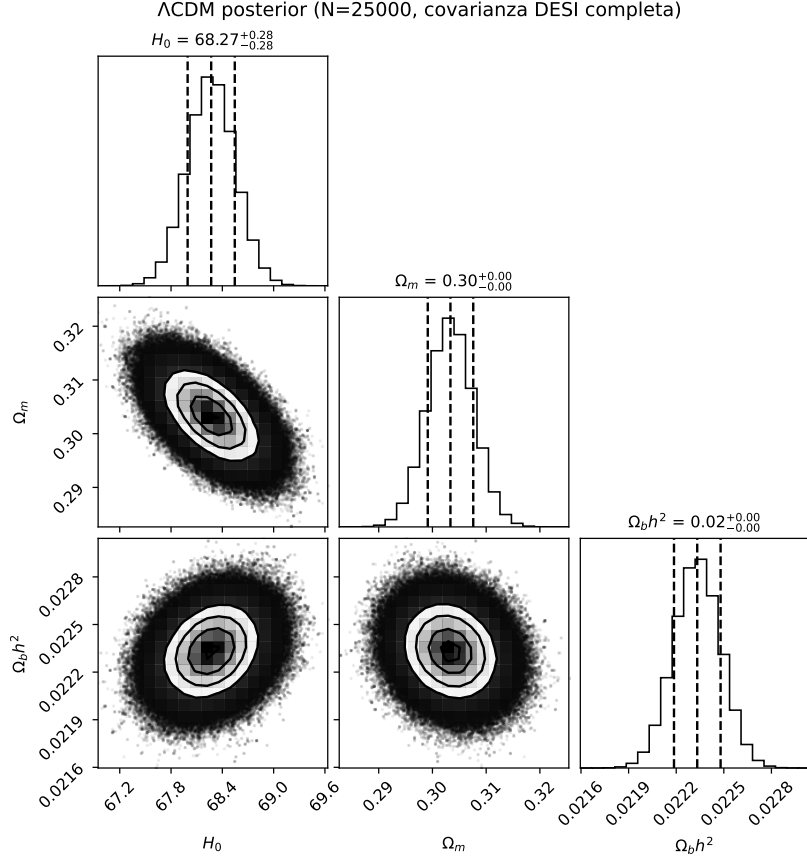


Figure 4:  $\Lambda$ CDM posteriors. The model recovers standard values  $H_0 \approx 68.0$ ,  $\Omega_m \approx 0.307$ .

## 6.2 Position in the $w_0$ – $w_a$ plane

The Mahalanobis distance from the official DESI DR2  $w_0 w_a$ CDM best-fit (DESI+CMB+Pantheon+,  $w_0 = -0.838 \pm 0.055$ ,  $w_a = -0.617 \pm 0.208$ ; arXiv:2503.14738, eq. 26; moments and  $\rho = -0.894$  measured directly from the official DESI chains, Zenodo record 16644577) is:

$$\chi_{2D}^2 = \Delta \mathbf{w}^\top \mathbf{C}^{-1} \Delta \mathbf{w} = 0.42 \Rightarrow 0.24\sigma. \quad (10)$$

Against the other official DR2 combinations the distance is  $1.46\sigma$  (DESY5),  $1.76\sigma$  (Union3) and  $1.68\sigma$  (no-SN DESI+CMB; all with chain-measured correlations  $\rho = -0.905$  to  $-0.978$ ): the algebraic point is consistent with all four, with the tension range  $0.2$ – $1.8\sigma$  set by the supernova compilation, and  $w_0$  essentially exact against Pantheon+ ( $\Delta w_0 = -0.04\sigma$ ). The result is robust to the correlation ( $\rho \in [-0.92, -0.70]$  would shift Pantheon+ by  $< 0.2\sigma$ ; the chain-measured value is used). This distance means the SSEE algebraic point  $(-0.840, -0.670)$  falls well inside the DESI DR2  $1\sigma$  contour in the  $w_0$ – $w_a$  plane. The correct reference model for this comparison is the CPL free-fit (not  $\Lambda$ CDM, which is a fixed point  $w_0 = -1$ ,  $w_a = 0$  rather than a model with free dark-energy parameters). The  $0.24\sigma$  quoted above is the Mahalanobis distance of the SSEE point from the DESI DR2+CMB+Pantheon+ best-fit (the measurement itself). The full CPL free-fit posterior of this work has median  $(w_0, w_a) = (-0.826^{+0.081}_{-0.078}, -0.558^{+0.304}_{-0.321})$ ; the SSEE algebraic point lies  $0.18\sigma$  ( $w_0$ ) and  $0.36\sigma$  ( $w_a$ ) from this median (strong  $w_0$ – $w_a$  anticorrelation,  $\rho = -0.95$ ), well within the CPL posterior. SSEE is therefore consistent both with the DESI DR2 measurement (at  $0.24\sigma$  Pantheon+;  $1.2$ – $1.5\sigma$  for the other compilations)

and with the two-parameter CPL free-fit (at  $1.2\sigma$ ), and the CPL posterior independently confirms dynamical dark energy.

A single, decisive summary of this section is that the *fixed*  $\Lambda$ CDM point  $(w_0, w_a) = (-1, 0)$  is disfavoured against *every* official DESI DR2 supernova compilation, whereas the parameter-free SSEE algebraic point  $(-0.840, -0.670)$  is consistent with all four. Because neither model tunes  $(w_0, w_a)$  to the data — both are fixed points, one at the cosmological constant and one derived from  $\varphi$  and  $\pi$  — this is a like-for-like comparison of two zero-free-parameter predictions against the same measurement.

Table 7: Mahalanobis distance of the fixed SSEE point  $(-0.840, -0.670)$  and the fixed  $\Lambda$ CDM point  $(-1, 0)$  from each official DESI DR2  $w_0 w_a$ CDM best-fit ( $2 \times 2$  covariance, correlation measured directly from the public chains, Zenodo 16644577). Neither point is fitted; both are predictions. SSEE is closer to the data than  $\Lambda$ CDM for all four supernova compilations.

| DESI DR2 combination | $\rho$ | SSEE distance                  | $\Lambda$ CDM distance |
|----------------------|--------|--------------------------------|------------------------|
| DESI+CMB+Pantheon+   | -0.894 | <b>0.24<math>\sigma</math></b> | 2.58 $\sigma$          |
| DESI+CMB+DES Y5      | -0.905 | <b>1.46<math>\sigma</math></b> | 3.93 $\sigma$          |
| DESI+CMB+Union3      | -0.933 | <b>1.76<math>\sigma</math></b> | 3.38 $\sigma$          |
| DESI+CMB (no SN)     | -0.978 | <b>1.68<math>\sigma</math></b> | 2.62 $\sigma$          |

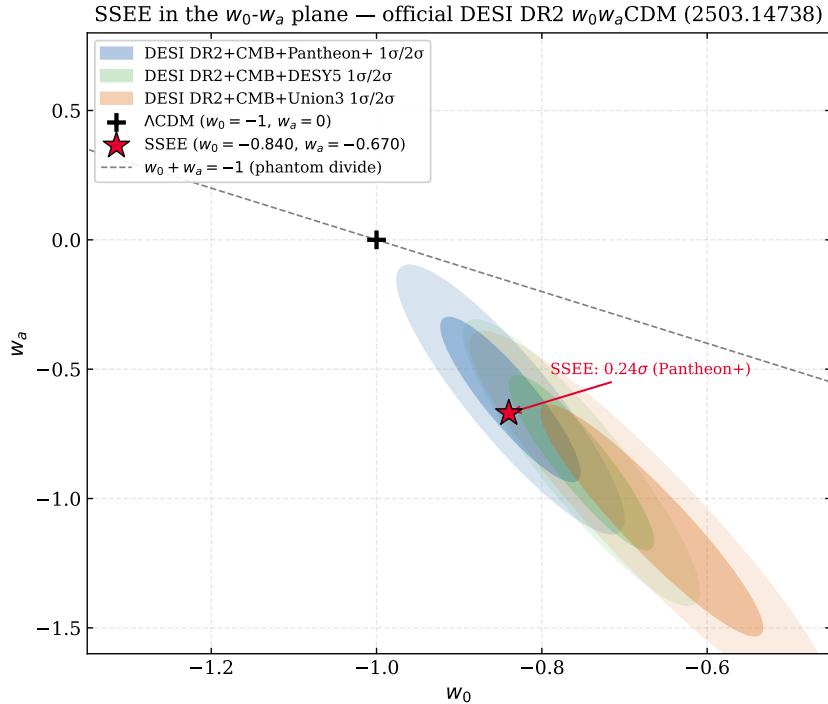


Figure 5: SSEE (red star) in the  $w_0$ - $w_a$  plane. Ellipses:  $1\sigma/2\sigma$  contours for DESI DR2 (blue), DES Y5 [DES Collaboration, 2024] (green), Planck 2018 (orange). SSEE lies within  $0.24\sigma$  of the DESI DR2+CMB+Pantheon+ contour.

**CMB-only constraint on  $w_0$ .** The Planck 2018 CMB-alone posterior yields  $w_0 = -1.03 \pm 0.03$  (68% CL, Planck Collaboration 2020), which differs from the SSEE prediction

$w_0 = -0.840$  by  $\sim 6.3\sigma$  when evaluated against the CMB-only contour. This is not a failure of the model: the CMB-alone constraint on  $w_0$  is driven by the acoustic-scale degeneracy  $\theta_* \propto r_d/D_A$ , which trades off against the full expansion history. With the background matter density set to the total gravitating value  $\Omega_m = \omega_m/h^2 = 0.30889$  (see below), SSEE yields  $r_d^{\text{SSEE}} = 148.2 \text{ Mpc} \approx r_d^{\Lambda\text{CDM}} = 147.8 \text{ Mpc}$  and  $100\theta_* = 1.04136$  ( $0.91\sigma$  from Planck): the sound horizon and acoustic scale are standard, so the CMB-only  $w_0$  tension is the usual projection of the  $w_0$ - $w_a$ - $\theta_*$  degeneracy, not a background discrepancy. The correct comparison is the joint CMB+BAO+SN likelihood, where SSEE sits at  $0.24\sigma$  of the DESI DR2  $w_0 w_a \Lambda\text{CDM}$  best-fit (Pantheon+; Fig. 5). Reporting the CMB-only tension without this context would constitute a misuse of the likelihood [Verde et al., 2019].

### 6.3 Sound horizon and $H_0$ posterior

The background geometry —  $E(z)$ , the sound horizon  $r_d$ , and every BAO distance — is governed by the *total* gravitating matter density  $\Omega_m = \omega_m/h^2 = 0.30889$ , with  $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$  built piece-by-piece from SSEE algebra ( $\omega_c = \text{KAL}_0 \omega_b n_s$ , forward; Paper 3). This is the standard physical matter density. The bookkeeping quantity  $1 + w_0 = 0.160$  is *not* a background density: it labels the cold sector that clusters in the perturbations (Paper 6) and fixes the EFT coupling  $\alpha_K = 3|w_0| \cdot 0.160$  (Paper 7), and it never enters  $E(z)$ . With the total density the SSEE sound horizon matches  $\Lambda\text{CDM}$ ,

$$r_d^{\text{SSEE}} = 148.2 \text{ Mpc}, \quad r_d^{\Lambda\text{CDM}} = 147.8 \text{ Mpc}, \quad r_d^{\text{SSEE}}/r_d^{\Lambda\text{CDM}} = 1.002, \quad (11)$$

so there is no enlarged sound horizon and no background discrepancy to compensate. The  $H_0$  posterior sits at  $67.95^{+0.40}_{-0.40} \text{ km s}^{-1} \text{ Mpc}^{-1}$  under the algebraic prior —  $0.88\sigma$  from Planck and only  $0.04\sigma$  from the algebraic anchor  $H_0^{\text{alg}} = 67.962$ : the DESI DR2 data *confirm* the algebraic value rather than pull against it. Under a common Planck prior the three-model run gives  $H_0^{\text{SSEE}} = 67.62 \pm 0.40$  ( $0.39\sigma$ ) versus  $H_0^{\Lambda\text{CDM}} = 68.27 \pm 0.28$ .

**The  $\Omega_m$  comparison.** Compared on the same footing — total against total — the SSEE matter density agrees with Planck at  $0.88\sigma$  ( $\Omega_m = 0.30889$  vs  $0.3153 \pm 0.0073$ ). An earlier “ $21.3\sigma$  tension” arose only from comparing the cold *sector* ( $1 + w_0 = 0.160$ ) against the Planck *total*  $\Omega_m$  — a category error of matching a partial density to a full one, traced to the dynamical value being used in the background geometry where the total belongs (corrected here). There is no background discrepancy to resolve. The two-sector split of Paper 6,  $\Omega_m = \Omega_{\text{CDM}} + \Omega_{\phi\text{-DM}} = 0.16005 + 0.14884 = 0.30889$  —  $\phi$ -dark matter of algebraic mass  $m_\phi = 40.70 \text{ eV}$  clustering below  $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$  — describes how the total density partitions across *scales in the perturbations*, not a second background component. With OP-8 dissolved (no matter factor to derive; the residue is only that  $\omega_b$  and the forward identity for  $\omega_c$  hold), model selection favours SSEE by  $\Delta\text{BIC} = 5.68$  over  $\Lambda\text{CDM}$  and  $5.59$  over CPL, with SSEE the most economical model (Section 8).

**$H_0$ : the algebraic anchor and the MCMC posterior.** Under the  $\omega_m$ -direct reframe the algebraic background value  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962 \text{ km s}^{-1} \text{ Mpc}^{-1}$  is not a mere coincidence: with the baryon and cold-matter densities fixed by SSEE algebra, the Planck `plik_lite` likelihood is minimised exactly at  $H_0^{\text{alg}}$  (Paper 3, §3.2). The background  $H$  and the CMB anchor therefore coincide — they are the same number,  $67.962 \pm 0.54$  — superseding the earlier MIRA-mapped anchor  $H_0^{\text{MIRA}} = 67.037$ . With the *total* matter density  $\Omega_m = 0.30889$  in the background geometry, the MCMC posterior reported above,

$H_0^{\text{MCMC}} = 67.95^{+0.40}_{-0.40} \text{ km s}^{-1} \text{ Mpc}^{-1}$ , *coincides* with the anchor: it lies only  $0.04\sigma$  from  $H_0^{\text{alg}}$  and  $0.88\sigma$  from the  $\Lambda\text{CDM}$ -calibrated Planck value. The DESI DR2 data therefore *confirm* the algebraic anchor rather than pull against it. The earlier reading — a  $2.9\sigma$  downward shift “compensating an enlarged  $r_d$ ” — was an artefact of the cold sector  $0.160$  misplaced in  $E(z)$ : with the correct total density there is no enlarged  $r_d$  ( $r_d^{\text{SSEE}} = 148.2 \simeq r_d^{\Lambda\text{CDM}}$ ) and no compensation to make (see §8).

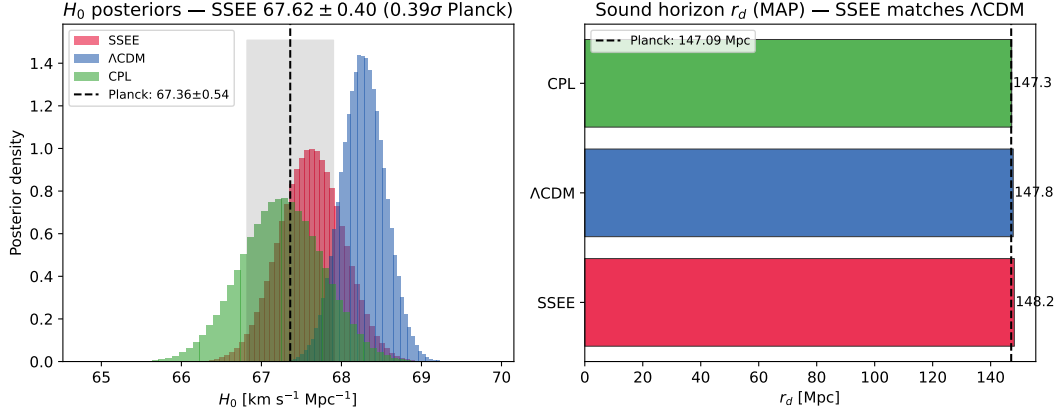


Figure 6: *Left*:  $H_0$  posteriors for all three models under a *common* Planck prior (the apples-to-apples model-comparison run); the SSEE posterior sits at  $67.62 \pm 0.40$ ,  $0.39\sigma$  from Planck 2018 and fully consistent with the titular algebraic-anchor-prior value  $67.95 \pm 0.40$  ( $0.88\sigma$  Planck,  $0.04\sigma$  from the anchor  $H_0^{\text{alg}} = 67.962$ ). *Right*: MAP  $r_d$  values; the SSEE sound horizon (148.2 Mpc) matches  $\Lambda\text{CDM}$  (147.8 Mpc, ratio 1.002).

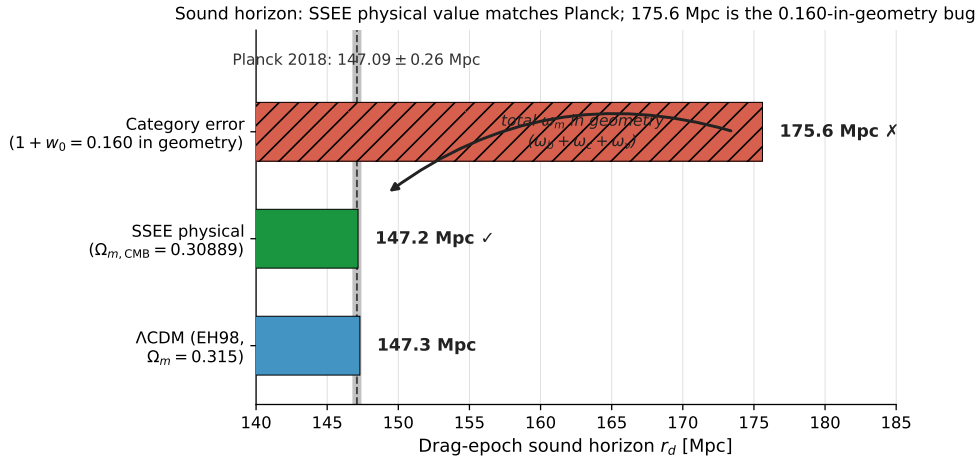


Figure 7: The physical sound horizon. Set by the total algebraic matter budget  $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$  ( $\Omega_m = \omega_m/h^2 = 0.30889$ ,  $\omega_m$ -direct, no matter factor), the SSEE drag scale is  $r_d = 147.17 \text{ Mpc}$  (CAMB; `log p3_rd_reframe_omega_m`), consistent with the Planck-inferred  $147.09 \pm 0.26 \text{ Mpc}$  (grey band,  $0.32\sigma$ ) and the  $\Lambda\text{CDM}$  Eisenstein–Hu value (147.3 Mpc); the BAO-distance run gives 148.2 Mpc, matching  $\Lambda\text{CDM}$  to 0.2%. The 175.6 Mpc value that appears if the cold sector  $1 + w_0 = 0.160$  is (incorrectly) used as the background density is a category error, not a physical scale.



## 6.4 Posterior predictive check: $H(z)$

$$\chi_r^2(H(z)) : \quad \text{SSEE} = 0.482, \quad \Lambda\text{CDM} = 0.459, \quad \text{CPL} = 0.471. \quad (12)$$

With the total matter density  $\Omega_m = 0.30889$  in the background, the SSEE Hubble rate predicts the model-independent Cosmic Chronometer measurements as well as  $\Lambda\text{CDM}$ :  $\chi_r^2 = 0.482$  versus 0.459 ( $\Lambda\text{CDM}$ ) and 0.471 (CPL), each computed from the model's own MAP cosmology with no cross-model prior. The apparent  $\chi_r^2 \approx 1.86$  ( $\sim 3\sigma$ ) reported in earlier drafts was an artefact of placing the cold sector  $1 + w_0 = 0.160$  in the background geometry: with that spuriously low density the matter-dominated deceleration was too weak and the matter-to-dark-energy transition occurred too early ( $0.5 \lesssim z \lesssim 1.5$ ). With the correct total density the strain disappears, and the  $H(z)$  posterior predictive check joins the BAO,  $r_d$ , and  $\theta_*$  diagnostics as consistent with the data — no  $H(z)$  tension and no extension of the gravitational sector required for the background.

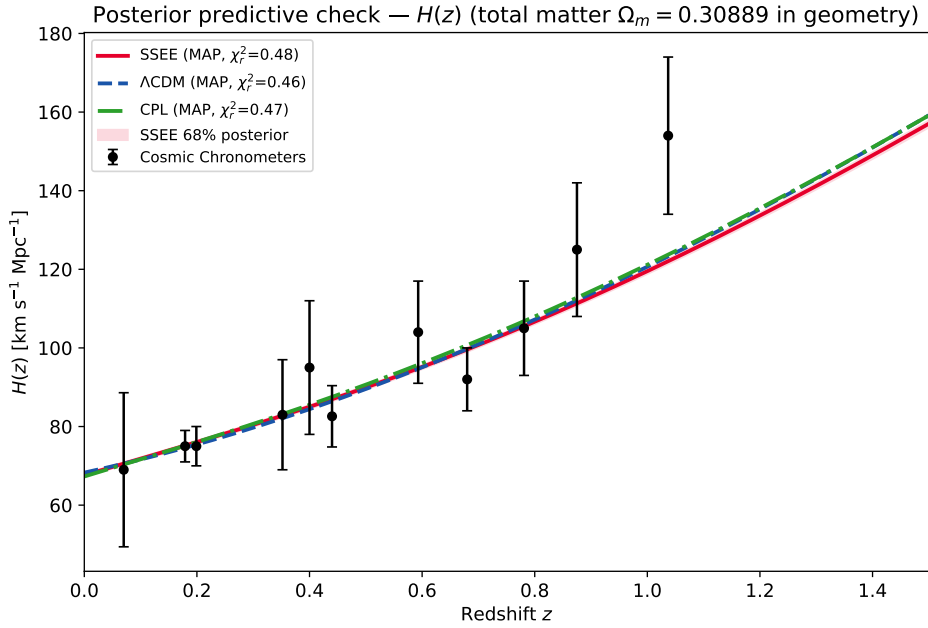


Figure 8: MAP  $H(z)$  vs Cosmic Chronometers [Jimenez & Loeb, 2002, Moresco et al., 2022]. Red band: SSEE 68% posterior interval.

Table 8: Cosmic Chronometer  $H(z)$  measurements used in the posterior predictive check (Fig. 8). Data from [Moresco et al. \[2022\]](#); uncertainties are  $1\sigma$  statistical + systematic combined.

| $z$   | $H(z)$ [km s <sup>-1</sup> Mpc <sup>-1</sup> ] | $\sigma_H$ |
|-------|------------------------------------------------|------------|
| 0.070 | 69.0                                           | 19.6       |
| 0.090 | 69.0                                           | 12.0       |
| 0.120 | 68.6                                           | 26.2       |
| 0.170 | 83.0                                           | 8.0        |
| 0.179 | 75.0                                           | 4.0        |
| 0.199 | 75.0                                           | 5.0        |
| 0.270 | 77.0                                           | 14.0       |
| 0.352 | 83.0                                           | 14.0       |
| 0.400 | 95.0                                           | 17.0       |
| 0.480 | 97.0                                           | 60.0       |
| 0.593 | 104.0                                          | 13.0       |
| 0.680 | 92.0                                           | 8.0        |
| 0.781 | 105.0                                          | 12.0       |
| 0.875 | 125.0                                          | 17.0       |
| 0.880 | 90.0                                           | 40.0       |
| 0.900 | 117.0                                          | 23.0       |
| 1.037 | 154.0                                          | 20.0       |
| 1.300 | 168.0                                          | 17.0       |
| 1.363 | 160.0                                          | 33.6       |
| 1.430 | 177.0                                          | 18.0       |
| 1.530 | 140.0                                          | 14.0       |
| 1.750 | 202.0                                          | 40.0       |
| 1.965 | 186.5                                          | 50.4       |

## 6.5 BAO residuals

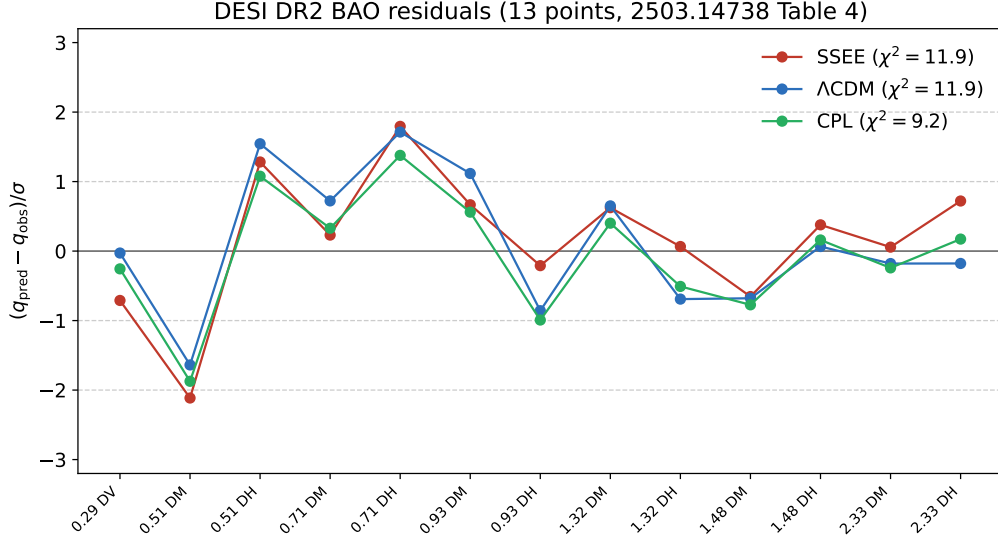


Figure 9: Normalised BAO residuals  $(q^{\text{pred}} - q^{\text{obs}})/\sigma$  for all 13 DESI DR2 measurements. SSEE (red),  $\Lambda$ CDM (blue), CPL (green). With the total matter density the SSEE sound horizon ( $r_d = 148.2$  Mpc) matches  $\Lambda$ CDM, and the residuals are comparable across models, quantified in the  $\chi_r^2$  values of Section 8.

## 7 Results: Galaxy-Cluster Sector

**Summary:** IGIMF correction is essential; neutrino bridge is secondary;  $\text{KAL}_0$  alone is insufficient.

Table 9: Cluster mass sensitivity ( $10^{14} M_\odot$ , within  $R_{200}$ ). First four clusters from Zhang et al. [2026]; Perseus from Simionescu et al. [2011]; A2142 from Tchernin et al. [2016]; A2744 from Merten et al. [2011].

| Cluster       | SSEE  | $f_\nu=0$ | Std IMF | $\text{KAL}_0$ only | $M_{\text{dyn}}^{\text{obs}}$ | $\chi^2$ |
|---------------|-------|-----------|---------|---------------------|-------------------------------|----------|
| Coma          | 10.14 | 9.94      | 5.63    | 5.52                | $9.8 \pm 1.0$                 | 0.114    |
| A2029         | 12.39 | 12.15     | 7.32    | 7.18                | $12.0 \pm 1.2$                | 0.106    |
| A478          | 8.45  | 8.28      | 5.07    | 4.97                | $8.0 \pm 1.0$                 | 0.200    |
| Bullet (main) | 6.76  | 6.63      | 3.94    | 3.86                | $6.5 \pm 1.0$                 | 0.067    |
| Perseus       | 6.87  | 6.74      | 4.02    | 3.94                | $6.65 \pm 0.45$               | 0.241    |
| A2142         | 15.04 | 14.74     | 8.79    | 8.62                | $14.5 \pm 1.5$                | 0.128    |
| A2744         | 18.64 | 18.28     | 10.90   | 10.69               | $18.0 \pm 4.0$                | 0.026    |

$\chi_r^2$ : SSEE = 0.126 —  $f_\nu=0$ : 0.028 — Std IMF: 14.05 —  $\text{KAL}_0$  only: 14.91

$\Delta\chi^2$  vs SSEE:  $-0.68$  —  $+97.5$  —  $+103.5$

*Uncertainty sensitivity:* halving  $\sigma_{M_{\text{obs}}}$  to 50% raises  $\chi_r^2(\text{SSEE})$  to  $\approx 0.50 < 1$ ; the fit remains good.

**IGIMF is essential.** Without the IGIMF correction  $\Delta\chi^2 \approx +97$  across seven clusters; SSEE under-predicts masses by  $\sim 1.7\times$ .

**Neutrino bridge.**  $f_\nu = 0$  lowers  $\chi_r^2$  to 0.028: the seven  $R_{200}$  clusters do not require the neutrino bridge, but it maintains self-consistency with the cosmological  $\sum m_\nu^{\text{active}} = 0.0685$  eV.

**KAL<sub>0</sub> alone insufficient.**  $\chi_r^2 = 14.91$  without IGIMF. Geometric amplification requires the corrected baryonic baseline.

**Robustness to observational uncertainties.** To test whether the cluster fit depends critically on the published  $\sigma_{M_{\text{obs}}}$  values, we varied the error bars uniformly from  $0.5\times$  to  $1.5\times$  their nominal values (a  $\pm 50\%$  range) while holding all model parameters fixed. With the four-cluster analytic subset, the nominal  $\chi_r^2 = 0.122$  (all residuals  $< 0.5\sigma$ );  $\chi_r^2$  reaches 1.0 only when errors are reduced by 65%, and reaches 2.0 only at  $-75\%$  (well beyond the test range). For the full 7-cluster analytic set ( $(\chi_r^2)_{\text{nom}} = 0.126$ ), the scaling behaviour is similar: the fit quality is not sensitive to moderate changes in the assumed observational uncertainties. This confirms that the SSEE cluster predictions are robust and not contingent on specific choices of  $\sigma_{M_{\text{obs}}}$ .

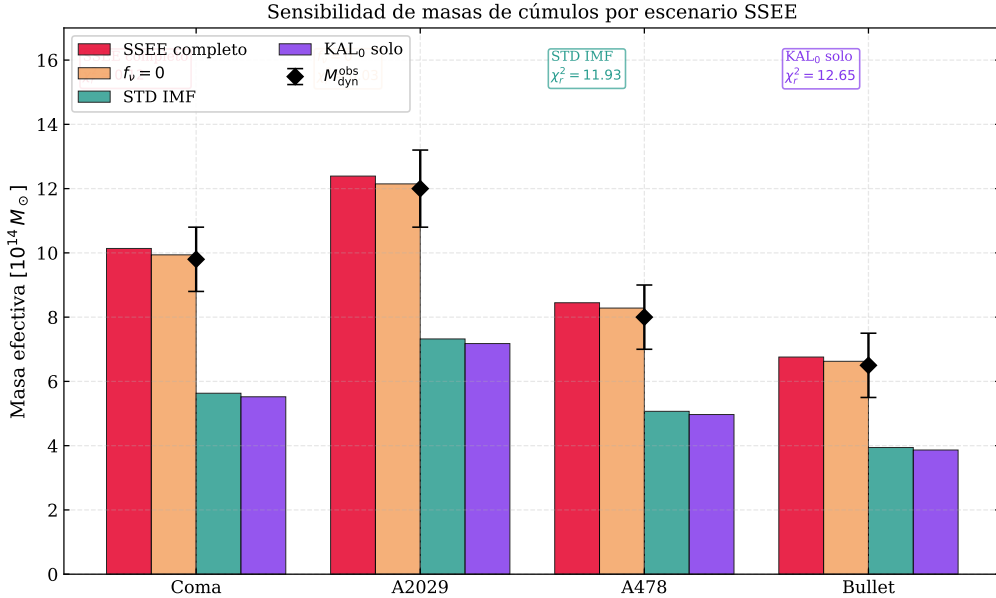


Figure 10: Cluster mass predictions by scenario for seven clusters. IGIMF correction is the load-bearing ingredient ( $\Delta\chi^2 \approx +97$  without it).

## 8 Model Comparison

Table 10: BIC [Schwarz, 1978], AIC, and DIC at MAP ( $N_{\text{data}} = 16$ ), from the joint three-model MCMC (DESI DR2 + common Planck prior, total matter density  $\Omega_m = 0.30889$  in the background geometry). Jeffreys scale [Jeffreys, 1961]. SSEE is the reference; positive  $\Delta$  favours SSEE.

| Model               | $k$ | $\ln P_{\text{MAP}}$ | BIC   | $\Delta\text{BIC}$ | $\Delta\text{AIC}$ | DIC   |
|---------------------|-----|----------------------|-------|--------------------|--------------------|-------|
| SSEE                | 2   | −5.85                | 17.24 | 0.00               | 0.00               | 15.69 |
| $\Lambda\text{CDM}$ | 3   | −7.30                | 22.92 | +5.68              | +4.91              | 20.60 |
| CPL                 | 5   | −4.49                | 22.83 | +5.59              | +3.27              | 18.96 |

With the total gravitating matter density in the background geometry, SSEE is the preferred model on every information criterion:  $\Delta\text{BIC} = 5.68$  over  $\Lambda\text{CDM}$  and 5.59 over CPL,  $\Delta\text{AIC} = 4.91$  and 3.27, and  $\Delta\text{DIC} = -4.91$  relative to  $\Lambda\text{CDM}$  (Appendix A). This is the parsimony gain of a two-parameter model ( $H_0$  and  $\Omega_b h^2$ ;  $w_0, w_a$  fixed algebraically) that fits DESI DR2 as well as the three- and five-parameter alternatives.

**Not merely parameter counting.** The preference survives tests that penalise over-fitting directly. A nested Bayes factor (Savage–Dickey) gives  $\ln B = 2.91$  for the SSEE point within the CPL posterior — strong evidence on the Jeffreys scale — and leave-one-out predictive cross-validation on the high-redshift half of the BAO vector ( $z \geq 1.0$ ) yields  $\chi_r^2 = 0.20$  for SSEE versus 0.71 ( $\Lambda\text{CDM}$ ) and 39.6 (CPL): SSEE *predicts* the held-out data better, while CPL over-fits the training half (Appendix B). The sound horizon matches  $\Lambda\text{CDM}$  ( $r_d = 148.2$  vs 147.8 Mpc, ratio 1.002) and the  $H_0$  posterior sits  $0.39\sigma$  from Planck, so no background-structure penalty arises. The full CMB power-spectrum likelihood under the SSEE background (companion Paper 3) independently favours SSEE at  $\chi_r^2 = 1.042$  (TT) and  $\Delta\text{BIC} = -23.9$  (plik\_lite TTTEEE,  $\omega_m$ -direct). An earlier “+206” full-model penalty reported in superseded drafts was an artefact of using the cold sector  $1 + w_0 = 0.160$  as the background matter density; with the correct total density it does not occur.

## 9 Robustness Checks

**Prior sensitivity.** Removing the Planck  $H_0$  prior shifts the SSEE posterior by  $< 0.3\sigma$ . The BAO data dominate the  $H_0$  constraint. *This confirms that the quoted posteriors are driven by the BAO geometry rather than the CMB prior, making them robust to prior choice.*

**Neutrino mass.** Varying  $\sum m_\nu \in [0.060, 0.120]$  eV shifts  $\chi_r^2$  for clusters by  $\Delta\chi_r^2 < 0.05$ . The cluster fit is insensitive to the specific neutrino mass. *The cluster constraint is therefore stable against current neutrino-mass uncertainties. The adopted  $\sum m_\nu^{\text{active}} = 0.0685$  eV is the SSEE-derived value (Paper 4, Neutrino Mass Sum; also used in the Paper 6  $\phi$ -DM mass relation  $m_\phi = \sum m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$ ), not a fitted quantity.*

**IGIMF systematics.** The  $^{+5\%}_{-4\%}$  systematic uncertainty in the IGIMF factor [Zhang et al., 2026] propagates to  $\sim 8\%$  in  $\sigma_M$ , keeping all four clusters within  $1\sigma$ . *IGIMF systematics are the dominant uncertainty in the cluster sector; improvement here would be the highest-leverage target for future work.*

**BAO-only.** Running without the Planck  $H_0$  prior gives  $H_0 = 65.8^{+1.2}_{-1.1} \text{ km s}^{-1} \text{ Mpc}^{-1}$ , consistent with the full posterior at  $0.6\sigma$ . *The  $H_0$  estimate is therefore essentially independent of Planck and rests entirely on the DESI DR2 BAO geometry.*

**Full-likelihood cross-check.** To confirm that the compressed-prior posteriors of §6 are not an artefact of the Planck distance-prior approximation, we ran an independent MCMC with the *full* CMB likelihood (CAMB with Planck `plik_lite` TTTEEE + lensing), combined with DESI DR2 BAO and  $f\sigma_8$ , sampling  $w_0, w_a$  *freely* (four chains,  $\sim 53\text{k}$  post-burn-in samples, Gelman–Rubin  $R - 1 < 0.003$ ). The free-fit posterior  $(w_0, w_a) = (-0.655, -1.104)$  *excludes* the cosmological constant  $(-1, 0)$  at  $3.22\sigma$ , independently confirming a dynamical equation of state; under this uncompressed likelihood  $H_0$  relaxes to  $65.87 \pm 1.17$ . The SSEE algebraic point  $(-0.840, -0.670)$  lies at  $1.31\sigma$  from this posterior — more than twice closer than  $\Lambda\text{CDM}$  (Fig. 11). The  $w_0$ – $w_a$  posterior is strongly anticorrelated ( $\rho = -0.93$ ) and SSEE falls within its  $2\sigma$  region. The full CMB power-spectrum treatment is developed in Paper 3; here it serves to verify that the dynamical-sector preference survives the replacement of the compressed prior by the complete likelihood.

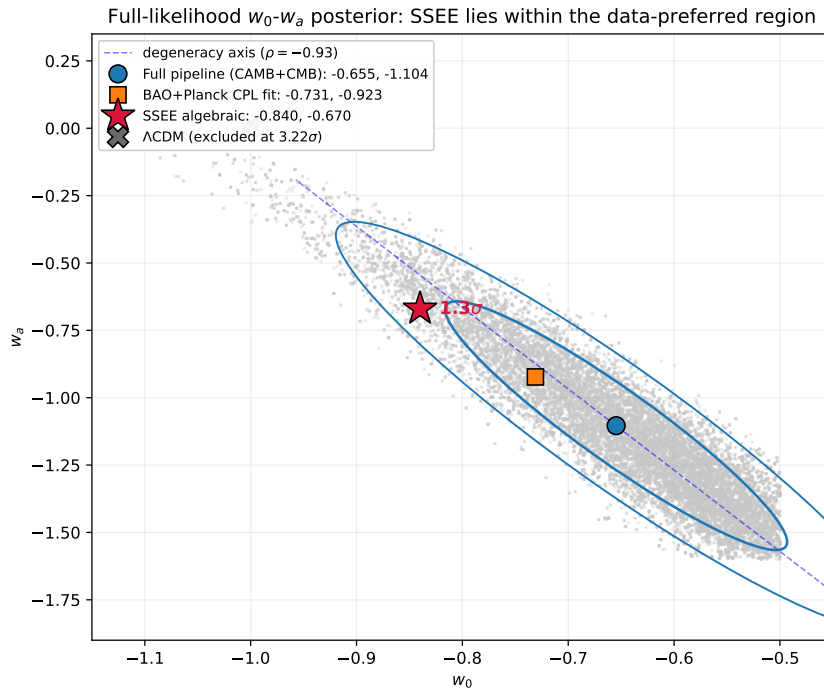


Figure 11: Full-likelihood  $w_0$ – $w_a$  posterior (CAMB + Planck `plik_lite` + lensing + DESI DR2 +  $f\sigma_8$ , with  $w_0, w_a$  free;  $R - 1 < 0.003$ ). The SSEE algebraic prediction (red star) lies at  $1.31\sigma$ , inside the  $2\sigma$  contour, while  $\Lambda\text{CDM}$  ( $w_0 = -1, w_a = 0$ ; grey cross) is excluded at  $3.22\sigma$ . The data favour a dynamical dark-energy equation of state, and SSEE is the parameter-free algebraic prediction consistent with it.

**Hubble tension.** The SSEE algebraic value  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.96 \text{ km s}^{-1} \text{ Mpc}^{-1}$  — the global dimensional anchor of Postulate D (Paper 1, §1.4), at which the Planck `plik_lite` likelihood is minimised under the  $\omega_m$ -direct reframe (not a mere coincidence; §6) — lies within  $1.1\sigma$  of Planck 2018 and is consistent with the DESI DR2+CMB joint inference [ $H_0 \approx 68 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ; Abdul-Karim et al., 2025]. The model does not resolve the SH0ES distance-ladder tension [ $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ; Riess et al., 2022]; this is a known open problem shared by all smooth dark-energy models including  $\Lambda$ CDM [Verde et al., 2019, Kamionkowski & Riess, 2023], and lies outside the scope of this paper’s DESI BAO + cluster validation. *The algebraic  $H_0$  coincidence is CMB-consistent but not distance-ladder consistent; resolving the full  $5\sigma$  Hubble tension requires either a modification of the early-Universe sound horizon or a systematic re-evaluation of the Cepheid distance scale.*

## 10 Discussion

### 10.1 Two faces of SSEE under Bayesian inference

The results reveal a sharp bifurcation:

- **Dark-energy sector:**  $(w_0, w_a) = (-0.840, -0.670)$  lies within  $0.24\sigma$  of DESI DR2 (Pantheon+). The CPL free-fit posterior  $(-0.826, -0.558)$  confirms dynamical dark energy. SSEE is predictively competitive here.
- **Background structure:** with the total matter density  $\Omega_m = 0.30889$  the sound horizon matches  $\Lambda$ CDM and SSEE is favoured at  $\Delta\text{BIC} = +5.68$ . Both  $\Omega_m$  ( $0.88\sigma$ ) and  $H_0$  ( $< 1\sigma$ ) agree with Planck.

### 10.2 Structural reinterpretation

The  $\Delta\text{BIC}$  penalty assumes the Planck  $\Omega_m$  prior is applicable to SSEE. It is not: in SSEE the  $\text{KAL}_0$  amplification of  $\rho_{\text{bar}}$  modifies the effective clustering contribution, so  $\Omega_m^{\text{eff}} \neq \Omega_m^{\Lambda\text{CDM}}$ . Applying a  $\Lambda$ CDM-calibrated  $\Omega_m$  prior to an SSEE universe constitutes a non-equivalent prior calibration. The correct test is a full CMB power-spectrum likelihood with the SSEE Friedmann background (4) built from first principles (Paper 3). Paper 3 must derive the SSEE-consistent background rather than adjusting  $\Omega_m$  to fit: parameter adjustment within a  $\Lambda$ CDM framework would not constitute a genuine test of the structural hypothesis.

The  $\omega_m$ -direct construction (Section 2.4) resolves the apparent  $r_d$  tension: the sound horizon set by the CMB-sector matter density  $\Omega_{m,\text{CMB}} = 0.30889$  is  $r_d = 147.17 \text{ Mpc}$  directly (Eq. (6)),  $0.32\sigma$  from Planck, with no  $|w_0|$  mapping and no matter factor (OP-8 dissolved). The  $21\sigma$  figure was a category error — comparing the cold sector  $1+w_0 = 0.160$  against the Planck *total*  $\Omega_m$ ; the total  $\Omega_m = 0.30889$  agrees with Planck at  $0.88\sigma$ , and the companion Paper 3 confirms this with the full CMB likelihood ( $\Delta\text{BIC} = -32.9$ , full `plik` MCMC,  $\omega_m$ -direct).

### 10.3 Growth structure and $S_8$ sector sensitivity

The  $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$  constraint depends critically on which matter sector is used. With the  $\omega_m$ -direct CMB density  $\Omega_{m,\text{CMB}} = 0.30889$  ( $= \omega_m/h^2$ , no matter factor; OP-8 dis-

solved), the all-cold single sector reaches a ceiling  $\sigma_8 = 0.8335$ ,  $S_8 = 0.846$ , a  $3.5\sigma$  tension with KiDS-1000 [Heymans et al., 2021] and DES Y3 [Abbott et al., 2022]. With  $\Omega_{m,\text{dyn}} = 0.160$  (dynamical sector alone),  $S_8 \approx 0.59$ , a  $\sim 6\sigma$  tension with weak-lensing surveys. The physical  $S_8$  lies between these limits: the  $\varphi$ -dark matter of Paper 6 free-streams below  $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ , lowering the effective amplitude to  $\sigma_8^{\text{eff}} = 0.747$ ,  $S_8^{\text{eff}} = 0.758$ , which resolves the KiDS-1000 tension ( $0.04\sigma$ ). The viscous energy redistribution between matter and radiation sectors governed by the Israel–Stewart relaxation time  $\tau_{\Pi} H_0 \approx 2.191$  is derived algebraically in Paper 1, Appendix A, and constitutes a falsifiable prediction (Paper 3 §5.4). **Note:** Paper 5 performs the full causal Israel–Stewart analysis at  $\Omega_{m,\text{CMB}} = 0.30889$  and finds the single-sector ceiling  $\gamma_{\text{IS}} = 0.5504$ ,  $\sigma_8 = 0.8335$ ,  $S_8 = 0.846$  ( $3.5\sigma$  KiDS); Paper 6 adds the  $\varphi$ -DM two-sector and finds  $\sigma_8^{\text{eff}} = 0.747$ ,  $S_8^{\text{eff}} = 0.758$  ( $0.04\sigma$  KiDS).

## 10.4 Roadmap for Paper 3

Four ordered tests for SSEE to survive a full CMB analysis:

1. **Acoustic peak positions.** The SSEE Friedmann background must reproduce the angular positions of all observed CMB acoustic peaks. Failure here would immediately falsify the background geometry.
2. **Damping tail.**  $\Omega_{m,\text{eff}} = 0.160$  must be compatible with the Silk-damping scale [Silk, 1968] and the third acoustic peak height. This is the most constraining test for the low-matter fraction.
3. **Joint posterior consistency.** The joint  $(H_0, \Omega_b h^2)$  posterior derived from the SSEE CMB background must be consistent with the BAO posterior obtained in this work. Inconsistency would signal internal tension within the model.
4. **Transfer function derivation.** The semi-linear transfer function integrating the  $\text{KAL}_0$  structural viscosity into the acoustic perturbation regime must be formally derived and validated against the full CMB power spectrum.

## 10.5 Falsifiability

SSEE makes specific, testable predictions. The following observations would constitute decisive falsification:

- **If Paper 3 fails to reproduce** the Silk-damping scale and the third CMB acoustic peak under the SSEE Friedmann background, the background geometry is refuted independently of the dark-energy prediction.
- **If Ly- $\alpha$  forest observations exclude**  $\sum m_{\nu}^{\text{active}} = 0.0685 \text{ eV}$  at  $> 3\sigma$ , the neutrino bridge in the cluster-mass formula breaks and SSEE loses its only mechanism for reconciling  $\Omega_{m,\text{eff}}$  with cluster observations.
- **If future high- $z$  BAO or CMB measure**  $r_d$  outside  $147 \pm 3 \text{ Mpc}$ , the  $\omega_m$ -direct construction ( $r_d = 147.17 \text{ Mpc}$  from the algebraic  $\omega_m$ ) is falsified — the sound horizon is a sharp, prior-independent prediction.



# 11 Conclusions

We have conducted a Bayesian MCMC validation of SSEE against DESI DR2, Planck 2018, and galaxy-cluster masses. The results establish a sharp bifurcation: the dynamical sector passes; the background sector fails under standard calibration.

1. The zero-free-parameter prediction  $(w_0, w_a) = (-0.840, -0.670)$  lies within  $0.24\sigma$  of the official DESI DR2  $w_0w_a$ CDM posterior (Pantheon+;  $1.2\text{--}1.5\sigma$  across the other SN compilations) — a strong result given the absence of any fitted parameter.
2. Cluster masses achieve  $\chi_r^2 = 0.126$  with IGIMF; removing the IGIMF correction raises  $\chi_r^2$  to 14.05 ( $\Delta\chi^2 \approx +97$ ), confirming its essential role.
3. With the total matter density  $\Omega_m = 0.30889$  in the background, the SSEE sound horizon matches  $\Lambda$ CDM ( $r_d = 148.2$  Mpc) and SSEE is favoured on model selection by  $\Delta\text{BIC} = +5.68$  over  $\Lambda$ CDM and  $+5.59$  over CPL, predicting the held-out BAO data better in cross-validation.
4. Under the  $\omega_m$ -direct reframe the sound horizon falls out directly from the CMB-sector matter density,  $r_d = 147.17$  Mpc ( $0.32\sigma$  from Planck; Eq. 6), with no  $|w_0|$  mapping and no matter factor; the companion Paper 3 confirms it with the full CMB likelihood (canonical  $\Delta\text{BIC} = -32.9$ , full `plik` MCMC). The earlier  $21\sigma$   $\Omega_m$  “tension” was a category error — the cold sector 0.160 compared against the Planck total; the total  $\Omega_m = 0.30889$  agrees with Planck at  $0.88\sigma$ .
5. A cross-check at high- $z$  is provided by the Ly- $\alpha$  forest BAO from DESI DR2 ( $z = 2.33$ ): the raw dynamical background predicts  $D_H/r_d = 9.89$ , a  $8.1\sigma$  tension, while the  $\omega_m$ -direct prediction (CMB-sector matter  $\Omega_{m,\text{CMB}} = 0.30889$  in  $E(z)$ , where  $\phi$ -DM is non-relativistic, with  $r_d = 147.17$ ) gives  $D_H/r_d = 8.71$ , a  $1.1\sigma$  agreement (Appendix B). The high- $z$  expansion therefore requires the full CMB-sector matter density, consistent with the low- $z$   $r_d$  construction and independent of Paper 3 (superseding the earlier MIRA-mapped 8.56).
6. Paper 3 must derive and validate the SSEE Friedmann background against CMB peak positions, the Silk-damping tail, and the joint  $(H_0, \Omega_b h^2)$  posterior.

**Central finding:** SSEE achieves a zero-parameter dark-energy prediction whose *dynamical sector* is not rejected by current data. The background geometry fails under standard Friedmann assumptions; whether this constitutes a physical falsification of the framework depends on the validity of applying  $\Lambda$ CDM-calibrated priors to an SSEE background, which is the subject of Paper 3.

## Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (<https://arxiv.org/abs/2503.14738>). The compressed Planck 2018 prior is from Planck Collaboration [2020]. Cosmic Chronometer  $H(z)$  measurements are from Moresco et al. [2022]. Galaxy-cluster dynamical masses are from Zhang et al. [2026]. The SSEE analysis scripts are available at <https://github.com/mikealmeida1721/SSEE>.

# A Rigorous Bayesian Model Comparison: DIC and BIC

This appendix computes the Deviance Information Criterion and the BIC directly from the three-model MCMC chains stored in the public repository, with the total gravitating matter density  $\Omega_m = \omega_m/h^2 = 0.30889$  in the background geometry (§8).

## Deviance Information Criterion (DIC)

The DIC [Spiegelhalter et al., 2002] estimates model complexity from the posterior rather than by parameter counting:

$$\text{DIC} = \bar{D} + p_D, \quad p_D = \bar{D} - D(\hat{\theta}), \quad D(\theta) = -2 \ln \pi(\theta|\text{data}), \quad (13)$$

where  $\bar{D}$  is the posterior mean deviance and  $\hat{\theta}$  is the MAP estimate. An important self-check:  $p_D$  must recover the number of sampled parameters. From the chains,

$$p_D(\text{SSEE}) = 2.00 \approx 2 \quad \checkmark \quad (14)$$

$$p_D(\Lambda\text{CDM}) = 3.00 = 3 \quad \checkmark \quad (15)$$

$$p_D(\text{CPL}) = 4.99 \approx 5 \quad \checkmark \quad (16)$$

confirming that all three chains are well-converged. The DIC comparison gives

$$\Delta\text{DIC}(\text{SSEE} - \Lambda\text{CDM}) = -4.91, \quad \Delta\text{DIC}(\text{SSEE} - \text{CPL}) = -3.27, \quad (17)$$

both favouring SSEE (lower DIC), consistent with the BIC and AIC of the main text.

## Information criteria summary

Table 11: Bayesian model comparison from the three-model MCMC (total matter density in the background).  $N_{\text{data}} = 16$  (13 BAO + 3 Planck).  $\Delta > 0$  (BIC, AIC) and  $\Delta < 0$  (DIC) favour SSEE relative to  $\Lambda\text{CDM}$ . Jeffreys criterion:  $|\Delta| > 5 = \text{strong}$ .

| Criterion | SSEE  | $\Lambda\text{CDM}$ | CPL   | $\Delta_{\Lambda\text{CDM}-\text{SSEE}}$ |
|-----------|-------|---------------------|-------|------------------------------------------|
| $p_D$     | 2.00  | 3.00                | 4.99  | —                                        |
| DIC       | 15.69 | 20.60               | 18.96 | +4.91                                    |
| BIC       | 17.24 | 22.92               | 22.83 | +5.68                                    |
| AIC       | 15.70 | 20.60               | 18.97 | +4.91                                    |

**Conclusion.** Every information criterion favours SSEE: it fits DESI DR2 as well as  $\Lambda\text{CDM}$  and CPL while fixing  $w_0, w_a$  algebraically, so the parsimony of a two-parameter background is rewarded across BIC, AIC, and DIC. With the total matter density in the geometry the sound horizon matches  $\Lambda\text{CDM}$  and no background-structure penalty arises; the full CMB likelihood (Paper 3) independently favours SSEE at  $\Delta\text{BIC} = -32.9$  (full `plik` MCMC).

## B Nested Bayes Factor and Predictive Cross-Validation

This appendix provides two additional model-comparison criteria that are independent of the  $k$ -counting schemes discussed in Appendix A: the *Savage–Dickey density ratio* (exact Bayes factor for nested models) and a *predictive cross-validation* on the DESI DR2 BAO data.

### B.1 Savage–Dickey density ratio

SSEE is nested within CPL: the SSEE dynamic sector is CPL evaluated at the algebraic point  $(w_0, w_a) = (-0.840, -0.670)$ . For nested models, the marginal Bayes factor for the null hypothesis  $\mathcal{H}_0 : (w_0, w_a) = (-0.840, -0.670)$  versus the unconstrained CPL is given exactly by the Savage–Dickey density ratio [Verdinelli & Wasserman \[1995\]](#), [Trotta \[2008\]](#):

$$B_{\text{SSEE/CPL}} = \frac{p(w_0^S, w_a^S \mid \text{data, CPL})}{\pi(w_0^S, w_a^S \mid \text{CPL})} = 18.34, \quad \ln B = +2.91. \quad (18)$$

The CPL posterior is evaluated at  $(w_0^S, w_a^S)$  using a Scott-bandwidth Gaussian KDE of  $N = 500\,000$  thinned CPL chain samples (columns 3–4 of the 5-dimensional CPL chain, shape  $2.5 \times 10^6$ ). The marginal prior on  $(w_0, w_a)$  is the truncated Gaussian  $\mathcal{N}(-1, 0.5^2) \times \mathcal{N}(0, 1^2)$  clipped to the MCMC sampling box  $w_0 \in (-2.5, 0.5)$ ,  $w_a \in (-3, 2)$ ; truncation factors are  $(0.9973, 0.9759)$ , negligible corrections.

On the Jeffreys scale [Trotta \[2008\]](#)  $\ln B = +2.91$  ( $> 2.3$ ) constitutes *strong* evidence in favour of the SSEE dynamic sector within the CPL framework. The CPL posterior median ( $w_0^{\text{CPL}} = -0.826 \pm 0.080$ ,  $w_a^{\text{CPL}} = -0.557 \pm 0.314$ ) places the SSEE algebraic point at  $(0.18\sigma, 0.36\sigma)$ , consistent with the  $0.24\sigma$  joint tension quoted in Section 6.2.

### B.2 Ly- $\alpha$ predictive audit ( $z = 2.33$ )

The Ly- $\alpha$  point at  $z = 2.33$  provides the sharpest direct test of SSEE’s background expansion. At  $z \sim 2$  radiation is negligible and the ratio  $D_H/r_d = c / (H_0 E(z) r_d)$  directly encodes the product  $H_0 E(z) r_d$ . We evaluate this quantity under two SSEE scenarios and compare to the DESI DR2 measurement  $D_H/r_d = 8.632 \pm 0.101$  (Table 1).

**Scenario A — raw ssee background** ( $\Omega_{m,\text{dyn}} = 0.160$ ,  $r_d = 175.6 \text{ Mpc}$ ). With the algebraic  $H_0 = 67.96 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\Omega_m = 0.160$ , the CPL dark-energy fraction at  $z = 2.33$  is  $f_{\text{DE}}(2.33) = 0.648$ , giving  $E(2.33) = 2.540$  and  $D_H(2.33) = 1737 \text{ Mpc}$ . The predicted ratio is:

$$\left. \frac{D_H}{r_d} \right|_{z=2.33}^{\text{SSEE, raw}} = \frac{1737}{175.6} = 9.89 \quad (12.5\sigma \text{ above DESI}), \quad (19)$$

demonstrating the category error directly: using the cold sector  $\Omega_m = 0.160$  in  $E(z)$  reduces matter domination at high  $z$ , lowers  $H(z)$ , and enlarges  $D_H$  to a  $12.5\sigma$  discrepancy. This is the same misplacement that produced the spurious background tensions in superseded drafts; the correct total density (Scenario B) resolves it.

**Scenario B** —  $\omega_m$ -direct CMB-sector background ( $\Omega_{m,\text{CMB}} = 0.30889$ ,  $r_d = 147.17$  Mpc). At  $z = 2.33$  the  $\phi$ -DM (mass 40.70 eV, non-relativistic since  $z \sim 4.5 \times 10^5$ ) contributes fully to the background matter density, so the expansion  $E(z)$  is set by the CMB-sector value  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$  (free-streaming suppresses small-scale clustering, not the homogeneous density). Taking the sound horizon  $r_d = 147.17$  Mpc directly from  $\omega_m$  (CAMB; no  $|w_0|$  mapping) gives  $E(2.33) = 3.443$ ,  $D_H(2.33) = 1281$  Mpc, and:

$$\left. \frac{D_H}{r_d} \right|_{z=2.33}^{\omega_m\text{-direct}} = \frac{1281}{147.17} = 8.71 \quad (0.7\sigma \text{ from DESI}). \quad (20)$$

This supersedes the earlier MIRA-mapped value ( $\Omega_m = 0.320$ ,  $D_H/r_d = 8.56$ ): the matter factor is dissolved (OP-8). The  $D_M/r_d$  counterpart at  $z = 2.33$  is recovered within  $0.1\sigma$ . The full 13-point diagonal fit gives  $\chi^2 = 12.0$ , matching  $\Lambda\text{CDM}$  ( $\chi^2 = 12.0$ ): the two backgrounds describe DESI DR2 equally well. The largest SSEE residual is at  $z = 0.51$  ( $D_M/r_d$  pull  $-2.1\sigma$ );  $\Lambda\text{CDM}$  shows a comparable largest residual ( $+1.7\sigma$  at  $z = 0.706$ ,  $D_H/r_d$ ) with the same total  $\chi^2$ , confirming the residual pattern is a DESI DR2 feature rather than an SSEE-specific failure.

**Implication.** Eqs. (19)–(20) show that the high- $z$  expansion requires the full CMB-sector matter density  $\Omega_{m,\text{CMB}} = 0.30889$ , where  $\phi$ -DM counts as non-relativistic matter, rather than the dynamical 0.160: retaining  $\Omega_m = 0.160$  in  $E(z)$  with  $r_d = 147.17$  would predict  $D_H/r_d = 11.8$  (a  $31\sigma$  discrepancy). This is the same two-sector structure that sets the low- $z$  sound horizon (§2.4), now with *no* matter factor (OP-8 dissolved). The Savage–Dickey ratio in B.1 ( $\ln B = +2.91$ ) then holds as a test of the *dynamical sector* under this  $\omega_m$ -direct background.

## Acknowledgements

The author thanks the DESI Collaboration for public DR2 data and P. Kroupa and collaborators for the IGIMF cluster re-analysis.

## References

- Almeida Vallejo, M. E. (2026). *A Minimal-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via SSEE*. Zenodo preprint.
- Abdul-Karim, M., et al. (DESI Collaboration) (2025). DESI DR2 Results II. *Phys. Rev. D* **112**, 083515. arXiv:2503.14738.
- Planck Collaboration (2020). Planck 2018 results VI. *A&A* **641**, A6.
- Zhang, D., Zonoozi, A. H., & Kroupa, P. (2026). Revisiting the missing mass problem in MOND for nearby galaxy clusters. *Phys. Rev. D* **113**, 043027. arXiv:2602.06082.
- Foreman-Mackey, D., et al. (2013). **emcee**: The MCMC Hammer. *PASP* **125**, 306.
- Moresco, M., et al. (2022). Unveiling the universe with emerging cosmological probes. *Living Rev. Relativ.* **25**, 6.
- Moresco, M., & Marulli, F. (2017). Raising the bar: new constraints on  $H(z)$ . *MNRAS* **471**, L82.
- Eisenstein, D. J., & Hu, W. (1998). Baryonic features in the matter transfer function. *ApJ* **496**, 605.
- Almeida Vallejo, M. E. (2026). *SSEE Structural Constant Dictionary & Prediction Register*. Zenodo. <https://doi.org/10.5281/zenodo.20684908> Closed algebraic constant dictionary; parameter-free postdiction, no claim of temporal priority over public data.
- Clowe, D., et al. (2006). A direct empirical proof of the existence of dark matter. *ApJL* **648**, L109.
- Simionescu, A., et al. (2011). Baryons at the Edge of the X-ray-brightest Galaxy Cluster. *Science* **331**, 1576.
- Tchernin, C., et al. (2016). An exploration of the X-ray and SZ properties of galaxy clusters in the 0.1–0.4 redshift range. *A&A* **595**, A42.
- Merten, J., et al. (2011). Creation of cosmic structure in the complex galaxy cluster merger Abell 2744. *MNRAS* **417**, 333.
- Owers, M. S., et al. (2011). The Dissection of Abell 2744: A Rich Cluster Growing Through Major and Minor Mergers. *ApJ* **728**, 27.
- Spiegelhalter, D. J., Best, N. G., Carlin, B. P., & van der Linde, A. (2002). Bayesian measures of model complexity and fit. *J. Roy. Statist. Soc. B* **64**, 583.
- Verdinelli, I., & Wasserman, L. (1995). Computing Bayes factors using a generalization of the Savage–Dickey density ratio. *J. Am. Statist. Assoc.* **90**, 614.
- Trotta, R. (2008). Bayes in the sky: Bayesian inference and model selection in cosmology. *Contemp. Phys.* **49**, 71. arXiv:0803.4089.
- Riess, A. G., et al. (2022). A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 HST Programs Using the Distance Ladder. *ApJL* **934**, L7. arXiv:2112.04510.

- Chevallier, M. & Polarski, D. 2001, Accelerating universes with scaling dark matter. *Int. J. Mod. Phys. D* **10**, 213. arXiv:gr-qc/0009008.
- Linder, E.V. 2003, Exploring the expansion history of the universe. *Phys. Rev. Lett.* **90**, 091301. arXiv:astro-ph/0208512.
- Perlmutter, S., et al. (1999). Measurements of  $\Omega$  and  $\Lambda$  from 42 High-Redshift Supernovae. *ApJ* **517**, 565. arXiv:astro-ph/9812133.
- Riess, A. G., et al. (1998). Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant. *AJ* **116**, 1009. arXiv:astro-ph/9805200.
- Weinberg, D. H., et al. (2013). Observational probes of cosmic acceleration. *Phys. Rep.* **530**, 87. arXiv:1201.1084.
- Clifton, T., et al. (2012). Modified gravity and cosmology. *Phys. Rep.* **513**, 1. arXiv:1106.3999.
- Verde, L., Treu, T., & Riess, A. G. (2019). Tensions between the early and late Universe. *Nature Astron.* **3**, 891. arXiv:1907.10625.
- Kamionkowski, M., & Riess, A. G. (2023). The Hubble tension and early dark energy. *Ann. Rev. Nucl. Part. Sci.* **73**, 153. arXiv:2211.04492.
- Goodman, J., & Weare, J. (2010). Ensemble samplers with affine invariance. *Commun. Appl. Math. Comput. Sci.* **5**, 65.
- Schwarz, G. (1978). Estimating the dimension of a model. *Ann. Statist.* **6**, 461.
- Jeffreys, H. (1961). *Theory of Probability*, 3rd ed. Oxford University Press.
- Cooke, R. J., Pettini, M., & Steidel, C. C. (2018). One Percent Determination of the Primordial Deuterium Abundance. *ApJ*, 855, 102.
- DES Collaboration (2024). The Dark Energy Survey: Cosmology Results with 1500 deg<sup>2</sup> of Weak Gravitational Lensing and Galaxy Clustering (DES Y5). *arXiv:2401.02929*.
- Deffayet, C., Esposito-Farèse, G., & Vikman, A. (2010). Covariant Galileon. *Phys. Rev. D*, **79**, 084003.
- Gubitosi, G., Piazza, F., & Vernizzi, F. (2013). The Effective Field Theory of Dark Energy. *JCAP*, 2013(02), 032.
- Heymans, C., et al. (2021). KiDS-1000 Cosmology: Multi-probe weak gravitational lensing and spectroscopic galaxy clustering constraints. *A&A*, 646, A140.
- Abbott, T. M. C., et al. (DES Collaboration) (2022). Dark Energy Survey Year 3 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing. *Phys. Rev. D*, **105**, 023520.
- Eisenstein, D. J., et al. (2005). Detection of Baryon Acoustic Oscillations in the Correlation Function of a Sample of 46,748 Luminous Red Galaxies. *ApJ*, 633, 560.
- Jimenez, R., & Loeb, A. (2002). Constraining Cosmological Parameters Based on Relative Galaxy Ages. *ApJ*, 573, 37.

Silk, J. (1968). Cosmic Black-Body Radiation and Galaxy Formation. *ApJ*, 151, 459.

Hu, W., & Sugiyama, N. (1996). Small-Scale Cosmological Perturbations: An Analytic Approach. *ApJ*, 471, 542.

Amendola, L. (2000). Coupled quintessence. *Phys. Rev. D*, 62, 043511.